

Sharing a Ride on the Commodities Roller Coaster: Common Factors in Business Cycles of Emerging Economies

Online Appendix*

This Draft: November 27, 2017

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This document presents the auxiliary material not presented in the main text of the paper. It is made of 8 sections. Section 1 reports further stylized facts. Section 2 reports the full set of equilibrium conditions, the formal definition of equilibrium and the measurement equations used when estimating the model. Section 3 reports further results related to the benchmark estimation not shown in draft. Section 4 reports further results related to the extensions and robustness described in the main draft. Sections 5 through 8 report results of three particular extensions using other detrending methods, using alternative dynamic factor models, and assuming non-stationary commodity prices. Section 9 discusses commodity production.

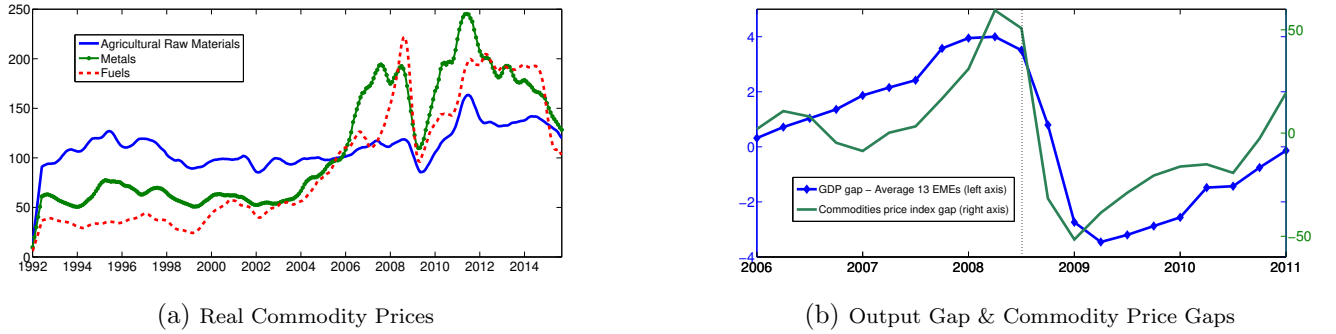
*Miguel Acosta, Javier Caicedo, Juan David Herreño, Santiago Téllez and Diego Zamora provided excellent research assistance. Any errors are our own. The information and opinions presented are entirely those of the authors, and no endorsement by the Inter-American Development Bank and Banco de la República, their Board of Executive Directors, or the countries they represent is expressed or implied. Corresponding author: andresf@iadb.org.

1. Further Stylized Facts

This subsection reports further stylized facts in addition to those reported in the main text. First, we present more descriptive statistics related to the linkages between commodity prices and economic activity (i) zooming in on the episode of the Great Recession; and (ii) on a longer time series range. Second, it presents the country-specific weights used for each of the commodity prices indices we construct and their stability through time. Third, we present the full set of business cycle (second) moments for both the larger sample of 13 EMEs as well as the 4 Latin American economies that we use when estimating the structural model. Fourth, we present more descriptive statistics on the comovement between commodity price indices across countries and goods, as well as in terms of economic activity. Fifth, we present the stylized facts documented in the main text with three alternative detrending filters. Sixth, we present descriptive statistics that document the size of the share of commodity exports in a typical EME. Seventh, we document the (little) trade that exists among the four Latin American economies that we study when estimating the structural model.

a. Commodity Prices and Real GDP

Figure A1. The Great Recession in Emerging Economies



Note: Left panel plots the six months centered moving average of commodity prices indices for three separate commodities: agricultural materials (solid/blue), metals (solid/green) and fuels (dashed/red) with 2005 as the base year (100). Right panel plots the quarterly cyclical components of a consolidated price index for all commodities (solid/green) and the simple average of the quarterly real GDP gap across 13 emerging economies (solid/blue). The dotted vertical line represents the quarter when Lehman declared bankruptcy. Cyclical components are computed with a Hodrick-Prescott filter with smoothing parameter of 1600. See text for the list of countries. Source: commodity prices come from IMF, real GDP are from Haver.

Figure A2. A Longer Period of Analysis

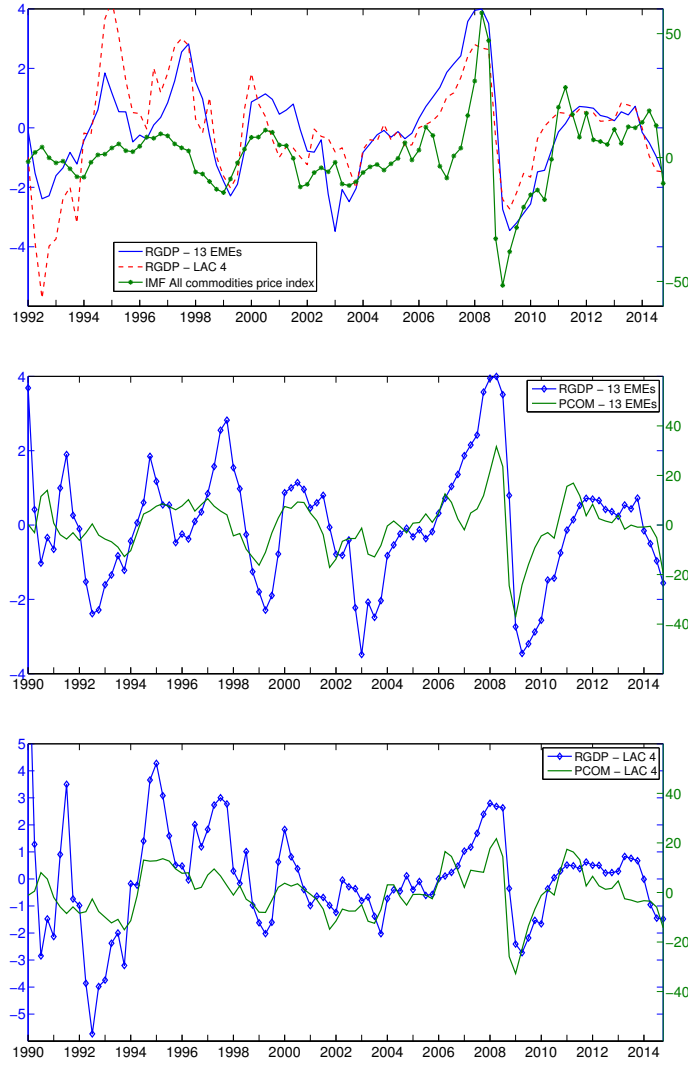


Table A1. Correlation coefficient table

correlation coefficient	P^{Co} IMF Index	P^{Co} EME 13 or LAC4
RGDP EME 13	0.644	0.477
RGDP LAC 4	0.621	0.522

Note: Cyclical average dynamics of the country-specific commodity indices full time series (left axis) and simple average real GDP cycle (right axis) for a pool of 13 EMEs and LAC4 between the first quarter of 1990 and the last quarter of 2014. Cyclical components were obtained with a Hodrick-Prescott filter (lambda of 1600). Upper panel reports the real GDP cycle for 13 EMEs and LAC4 (left axis) and the cyclical component of the IMF commodity price index. Middle panel shows an extended version of Figure 1.B with data from 1990. Lower panel reports simple average real GDP cycle for LAC4 instead of for 13 EMEs. Finally, a table with the correlation between the cyclical component of the RGDP for 13 EMEs and LAC4 with the IMF commodity index and the average country-specific commodity indices for EME 13 or LAC4, respectively, is reported.

b. Commodity Weights for Country-Specific Commodity Price Indices

Table A2. Weights (Thetas) by Commodity: All 13 EMEs & 44 commodities

	Argentina	Brazil	Bulgaria	Chile	Colombia	Ecuador	Malaysia	Mexico	Peru	Russian Federation	South Africa	Ukraine	Venezuela
Beef	2.6%	3.3%	0.0%	0.0%	0.1%	0.0%	0.0%	0.1%	0.0%	0.0%	0.2%	5.1%	0.0%
Lamb	0.0%	0.0%	2.1%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Pork	0.0%	1.4%	0.0%	0.8%	0.0%	0.0%	0.0%	0.9%	0.0%	0.0%	0.0%	0.2%	0.0%
Chicken	0.1%	5.9%	1.8%	0.3%	0.0%	0.1%	0.1%	0.0%	0.0%	0.0%	0.2%	0.0%	0.0%
Fish	2.4%	0.3%	0.1%	12.4%	0.8%	2.2%	0.3%	0.6%	1.1%	0.5%	3.6%	0.7%	0.1%
Wheat	7.8%	0.0%	5.9%	0.0%	0.0%	0.0%	0.2%	0.5%	0.0%	0.5%	0.7%	9.2%	0.0%
Rice	0.6%	0.0%	0.0%	0.0%	0.0%	0.4%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	0.0%
Barley	0.1%	0.0%	1.5%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.0%	4.3%	0.0%
Corn (Maize)	6.6%	1.0%	1.5%	0.7%	0.0%	0.3%	0.0%	0.1%	0.1%	0.0%	2.3%	0.9%	0.1%
Bananas	0.0%	0.1%	0.0%	0.0%	6.6%	25.1%	0.1%	0.1%	0.1%	0.0%	0.0%	0.0%	0.1%
Sugar	0.3%	9.0%	0.5%	0.0%	3.0%	0.4%	0.5%	0.4%	0.3%	0.1%	4.6%	0.5%	0.0%
Coffee	0.0%	8.5%	0.1%	0.1%	15.0%	1.5%	0.3%	2.7%	4.4%	0.0%	0.1%	0.0%	0.1%
Cocoa	0.0%	0.6%	0.3%	0.0%	0.1%	2.8%	1.2%	0.1%	0.3%	0.0%	0.0%	0.0%	0.0%
Tea	0.4%	0.2%	0.1%	0.1%	0.0%	0.0%	0.0%	0.1%	0.0%	0.0%	0.3%	0.0%	0.0%
Soybean Meal	16.0%	9.3%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Fish Meal	0.1%	0.0%	0.0%	3.1%	0.0%	0.4%	0.1%	0.0%	15.5%	0.0%	0.1%	0.0%	0.0%
Hides	5.4%	4.3%	2.2%	0.3%	1.4%	0.1%	0.3%	1.9%	0.2%	0.3%	3.7%	3.3%	0.1%
Groundnuts	0.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.4%	0.0%	0.0%
Soybeans	7.0%	12.5%	0.0%	0.0%	0.0%	0.3%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Natural Rubber	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	4.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Hard Log	0.0%	0.0%	0.3%	0.2%	0.0%	0.1%	3.6%	0.0%	0.0%	0.5%	0.2%	0.9%	0.0%
Soft Log	0.0%	0.1%	0.3%	0.2%	0.0%	0.0%	0.1%	0.0%	0.0%	1.9%	0.0%	1.1%	0.0%
Hard Sawn	0.0%	1.9%	1.8%	0.2%	0.0%	0.5%	5.8%	0.1%	1.3%	0.1%	0.1%	1.4%	0.0%
Soft Sawn	0.2%	1.2%	1.7%	5.4%	0.0%	0.0%	0.1%	0.6%	0.0%	1.3%	0.2%	2.1%	0.0%
Cotton	0.5%	0.4%	0.1%	0.0%	0.0%	0.0%	0.0%	0.3%	0.1%	0.0%	0.1%	0.0%	0.0%
Wool	0.3%	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	1.2%	0.0%	0.0%
Iron	0.0%	16.5%	0.0%	1.4%	0.0%	0.0%	0.0%	0.1%	1.6%	0.9%	8.9%	14.5%	0.4%
Copper	3.0%	0.6%	29.7%	69.6%	0.1%	0.0%	2.7%	3.3%	21.9%	1.8%	2.5%	3.6%	0.0%
Nickel	0.0%	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.6%	1.7%	0.1%	0.0%
Aluminum	2.2%	7.6%	2.6%	0.1%	0.5%	0.2%	1.5%	0.5%	0.0%	6.9%	16.5%	17.5%	3.8%
Lead	0.1%	0.0%	3.1%	0.0%	0.0%	0.0%	0.1%	0.1%	3.6%	0.0%	0.5%	0.1%	0.0%
Zinc	0.1%	0.2%	6.1%	0.1%	0.0%	0.0%	0.1%	1.3%	11.1%	0.2%	0.4%	0.0%	0.0%
Tin	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	0.9%	0.0%	1.2%	0.0%	0.0%	0.0%	0.0%
Coal	0.0%	0.0%	0.4%	0.0%	15.4%	0.0%	0.0%	0.0%	0.0%	1.9%	29.4%	4.0%	0.9%

Table A2. Weights (Thetas) by Commodity: All 13 EMEs & 44 commodities (cont.)

	Argentina	Brazil	Bulgaria	Chile	Colombia	Ecuador	Malaysia	Mexico	Peru	Russian Federation	South Africa	Ukraine	Venezuela
Soy Oil	8.8%	3.2%	0.0%	0.0%	0.0%	0.1%	0.5%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Olive Oil	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Sun Oil	4.1%	0.0%	1.2%	0.0%	0.0%	0.0%	0.1%	0.1%	0.0%	0.1%	0.2%	6.2%	0.0%
Palm Oil	0.0%	0.0%	0.0%	0.0%	0.5%	0.4%	20.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Coconut Oil	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Gold	0.7%	1.7%	0.0%	2.7%	2.1%	0.3%	0.7%	0.6%	28.0%	0.0%	0.9%	0.0%	0.2%
Shrimp	1.8%	0.6%	0.0%	0.0%	1.1%	9.2%	1.0%	2.3%	0.3%	0.0%	0.0%	0.0%	0.3%
Uranium	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%
Crude Oil	24.1%	8.6%	36.5%	2.1%	52.8%	55.5%	33.3%	82.5%	8.7%	54.8%	20.4%	20.6%	93.7%
NatGas	3.7%	0.0%	0.0%	0.2%	0.2%	0.0%	20.9%	0.5%	0.0%	25.5%	0.2%	3.6%	0.0%

Note: This table shows the weight of each commodity (rows) on the price index by country (columns). This is expressed in percentages (%).

Table A3. Stability of the Weights: LAC4 economies for 1999-2014

	1999	2000	2001	2002	2003	2004	2005	2006
Brazil	100.00%	94.13%	89.05%	83.36%	78.95%	79.23%	77.77%	70.69%
Chile	100.00%	99.98%	99.98%	99.97%	99.96%	99.88%	99.87%	99.78%
Colombia	100.00%	99.15%	97.94%	98.84%	94.38%	94.40%	95.02%	94.62%
Peru	100.00%	97.47%	97.99%	98.73%	97.92%	98.54%	98.54%	96.58%
	2007	2008	2009	2010	2011	2012	2013	2014
Brazil	63.98%	57.60%	70.05%	61.36%	61.44%	57.95%	64.24%	61.29%
Chile	99.77%	99.81%	99.81%	99.76%	99.76%	99.78%	99.82%	99.85%
Colombia	94.70%	94.78%	90.75%	94.32%	94.61%	93.67%	94.01%	95.04%
Peru	97.62%	98.07%	95.02%	96.24%	94.33%	93.02%	93.01%	96.12%

Note: This table depicts the evolution of Theta (commodity relative weight) over 1999-2014. For each country i and year t , the $i, j - th$ cell contains the correlation coefficient of $\theta_{i,1999}$ and $\theta_{i,t}$. Where $\theta_{i,t}$ is a 44x1 column vector, in which each element is the relative weight of each commodity in regards to the 44 commodities. Please note that $\theta_{i,t}\mathbf{1}' = 1$, where $\mathbf{1}'$ is a row vector of ones.

c. Business Cycle Moments

Table A4. All EMEs-13 & LAC4

	Latin America and the Caribbean (LAC 4)												
	S.D. (%)	Relative SD to RGDP	Autocorrelations		Cross-correlation with $RGDP_{t+j}$								
			1	2	$j = -4$	$j = -3$	$j = -2$	$j = -1$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$
P^{Co}	11.93	7.76	0.75	0.38	0.19	0.28	0.43	0.56	0.54	0.55	0.44	0.33	0.25
$RGDP$	1.54	1.00	0.76	0.46	-0.03	0.18	0.46	0.76	1.00	0.76	0.46	0.18	-0.03
GDP/CPI	2.73	1.73	0.73	0.47	0.02	0.19	0.40	0.58	0.72	0.69	0.56	0.40	0.27
$RGDP-G$	1.90	1.23	0.77	0.47	-0.09	0.14	0.43	0.74	0.97	0.76	0.48	0.22	0.01
$GDP-G/CPI$	3.11	1.98	0.73	0.47	-0.03	0.17	0.39	0.59	0.74	0.70	0.55	0.39	0.26
$Cons$	1.31	0.84	0.77	0.53	0.12	0.31	0.48	0.66	0.79	0.72	0.54	0.29	0.06
Inv	5.09	3.29	0.69	0.38	-0.02	0.14	0.36	0.63	0.82	0.66	0.35	0.11	-0.07
TB/GDP	1.81	1.13	0.62	0.43	0.05	0.06	0.02	-0.04	-0.02	0.04	0.13	0.22	0.27
$EMBIG$	1.12	0.77	0.57	0.18	-0.17	-0.13	-0.12	-0.08	-0.05	-0.17	-0.27	-0.32	-0.38
$CEMBI$	0.99	0.69	0.55	0.02	0.22	0.30	0.30	0.26	0.13	0.00	-0.10	-0.20	-0.18
RER	5.05	3.41	0.66	0.26	-0.08	-0.18	-0.28	-0.33	-0.26	-0.24	-0.21	-0.17	-0.13
Emerging Market Economies (EME 13)													
P^{Co}	12.98	6.74	0.72	0.30	0.11	0.25	0.42	0.56	0.53	0.53	0.38	0.22	0.10
$RGDP$	2.29	1.00	0.81	0.55	0.06	0.29	0.55	0.81	1.00	0.81	0.55	0.29	0.06
GDP/CPI	3.35	1.69	0.76	0.49	0.03	0.21	0.42	0.62	0.75	0.68	0.50	0.30	0.11
$RGDP-G$	2.75	1.37	0.80	0.52	-0.03	0.20	0.47	0.75	0.94	0.79	0.55	0.30	0.07
$GDP-G/CPI$	4.16	2.09	0.74	0.47	0.01	0.19	0.40	0.61	0.75	0.69	0.51	0.31	0.12
$Cons$	2.11	1.00	0.79	0.57	0.17	0.35	0.52	0.69	0.78	0.68	0.50	0.28	0.06
Inv	7.22	3.18	0.79	0.54	0.19	0.37	0.56	0.72	0.79	0.68	0.46	0.23	0.04
TB/GDP	2.31	1.20	0.55	0.35	-0.06	-0.11	-0.17	-0.19	-0.16	-0.13	-0.08	0.00	0.04
$EMBIG$	2.51	1.13	0.63	0.26	-0.04	-0.07	-0.10	-0.13	-0.13	-0.21	-0.23	-0.23	-0.24
$CEMBI$	1.84	0.73	0.63	0.18	0.27	0.22	0.12	-0.01	-0.17	-0.24	-0.25	-0.24	-0.19
RER	7.10	3.56	0.74	0.42	-0.01	-0.05	-0.14	-0.21	-0.23	-0.23	-0.19	-0.11	-0.07

Note: This table shows the standard deviation (first column), standard deviation relative to RGDP (second column), first and second autocorrelations (third and fourth column). and the cross-correlation with RGDP in $t = -4, \dots, 4$ for two country groups. LAC4 is composed by: Brazil, Colombia, Chile and Peru. EME13 is composed by LAC4, Bulgaria, Ecuador, Venezuela, Malaysia, Ukraine, Mexico, Argentina, South Africa and the Russian Federation. Data are filtered with Hodrick-Prescott filter (lambda=1600). Calculations were made with a sample from 2000.Q1 to 2014.Q4.

Table A5. All EMEs-13 & LAC4 (Extended sample)

	Latin America and the Caribbean (LAC 4)												
	S.D. (%)	Relative SD to RGDP	Autocorrelations		Cross-correlation with $RGDP_{t+j}$								
			1	2	$j = -4$	$j = -3$	$j = -2$	$j = -1$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$
P^{Co}	10.78	6.10	0.77	0.44	0.25	0.31	0.42	0.52	0.52	0.51	0.41	0.32	0.24
$RGDP$	1.95	1.00	0.67	0.39	0.05	0.23	0.39	0.67	1.00	0.67	0.39	0.23	0.05
GDP/CPI	2.73	1.46	0.71	0.43	0.10	0.28	0.46	0.62	0.73	0.67	0.53	0.38	0.22
$RGDP-G$	2.21	1.18	0.71	0.42	0.02	0.19	0.43	0.69	0.92	0.73	0.47	0.22	0.01
$GDP-G/CPI$	3.07	1.65	0.70	0.45	0.04	0.25	0.45	0.62	0.74	0.68	0.53	0.37	0.21
$Cons$	1.97	1.00	0.71	0.38	0.16	0.31	0.40	0.69	0.86	0.64	0.37	0.22	0.07
Inv	6.07	3.17	0.69	0.37	0.03	0.18	0.36	0.63	0.80	0.60	0.30	0.14	0.00
TB/GDP	1.93	1.04	0.56	0.23	0.12	0.08	0.06	0.03	0.07	0.14	0.16	0.15	0.16
$EMBIG$	1.22	0.68	0.55	0.15	-0.18	-0.15	-0.16	-0.13	-0.09	-0.18	-0.28	-0.30	-0.32
$CEMBI$	0.99	0.53	0.55	0.02	0.22	0.30	0.30	0.26	0.13	0.00	-0.10	-0.20	-0.18
RER	5.92	3.37	0.64	0.33	-0.11	-0.28	-0.35	-0.40	-0.36	-0.31	-0.27	-0.19	-0.12
Emerging Market Economies (EME 13)													
P^{Co}	11.71	5.53	0.73	0.35	0.13	0.25	0.39	0.51	0.49	0.47	0.33	0.20	0.09
$RGDP$	2.43	1.00	0.78	0.52	0.06	0.30	0.52	0.78	1.00	0.78	0.52	0.30	0.06
GDP/CPI	4.18	1.93	0.73	0.44	0.06	0.22	0.42	0.60	0.73	0.65	0.46	0.27	0.06
$RGDP-G$	2.92	1.32	0.77	0.48	-0.01	0.20	0.46	0.73	0.92	0.76	0.53	0.29	0.07
$GDP-G/CPI$	4.88	2.24	0.71	0.42	0.02	0.20	0.40	0.60	0.74	0.66	0.48	0.29	0.09
$Cons$	2.36	1.04	0.75	0.49	0.17	0.34	0.48	0.68	0.78	0.62	0.39	0.20	0.00
Inv	7.86	3.24	0.78	0.53	0.18	0.37	0.54	0.72	0.79	0.68	0.46	0.25	0.07
TB/GDP	2.83	1.35	0.56	0.33	-0.01	-0.08	-0.15	-0.19	-0.19	-0.17	-0.12	-0.06	-0.01
$EMBIG$	2.96	1.23	0.63	0.26	-0.08	-0.13	-0.20	-0.23	-0.22	-0.31	-0.31	-0.26	-0.21
$CEMBI$	1.84	0.65	0.63	0.18	0.27	0.22	0.12	-0.01	-0.17	-0.24	-0.25	-0.24	-0.19
RER	8.96	3.90	0.73	0.44	0.02	-0.06	-0.16	-0.25	-0.29	-0.30	-0.26	-0.18	-0.11

Note: This table shows the standard deviation (first column), standard deviation relative to RGDP (second column), first and second autocorrelations (third and fourth column). and the cross-correlation with RGDP in $t = -4, \dots, 4$ for two country groups. LAC4 is composed by: Brazil, Colombia, Chile and Peru. EME13 is composed by LAC4, Bulgaria, Ecuador, Venezuela, Malaysia, Ukraine, Mexico, Argentina, South Africa and the Russian Federation. Data are filtered with Hodrick-Prescott filter (lambda=1600). Calculations were made with a sample from 1990.Q1 to 2014.Q4.

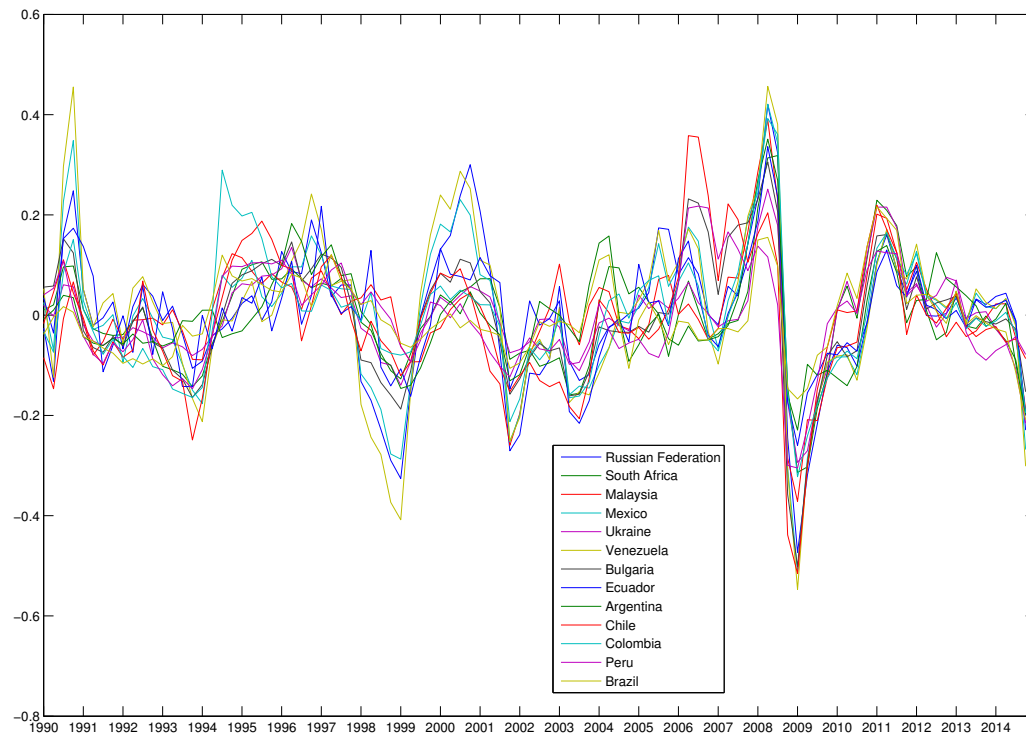
Table A6. Range of Time Series Data Coverage: EME 13

	Chile	Colombia	Peru	Brazil	South Africa	Russia	Argentina	Malaysia	Ukraine	Mexico	Ecuador	Venezuela	Bulgaria
P^{Co}	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4
$RGDP$	2003.Q1- 2014.Q4	2000.Q1- 2014.Q4	1990.Q1- 2014.Q4	1990.Q1- 2014.Q3	1990.Q1- 2014.Q4	1995.Q1- 2014.Q3	2004.Q1- 2014.Q3	2005.Q1- 2014.Q4	2001.Q1- 2014.Q4	1990.Q1- 2014.Q4	1990.Q1- 2014.Q3	1993.Q1- 2013.Q3	2000.Q1- 2014.Q4
GDP/CPI	1990.Q1- 2014.Q4	1994.Q1- 2014.Q3	1991.Q1- 2014.Q4	1995.Q1- 2014.Q3	1990.Q1- 2014.Q4	1993.Q1- 2014.Q4	2004.Q1- 2013.Q4	1991.Q1- 2014.Q4	2001.Q1- 2014.Q4	1990.Q1- 2014.Q4	2000.Q1- 2014.Q3	NA	1995.Q1- 2014.Q4
$GDP-G$	2003.Q1- 2014.Q4	2000.Q1- 2014.Q3	1991.Q1- 2014.Q4	1994.Q1- 2014.Q3	1990.Q1- 2014.Q4	1995.Q1- 2014.Q3	2004.Q1- 2014.Q3	2005.Q1- 2014.Q4	2001.Q1- 2014.Q4	1993.Q1- 2014.Q4	2000.Q1- 2014.Q3	NA	2000.Q1- 2014.Q4
$GDP-G/CPI$	1996.Q1- 2014.Q4	2000.Q1- 2014.Q3	1991.Q1- 2014.Q4	1995.Q1- 2014.Q3	1990.Q1- 2014.Q4	1993.Q1- 2014.Q4	2004.Q1- 2013.Q4	1991.Q1- 2014.Q4	2001.Q1- 2014.Q4	1993.Q1- 2014.Q4	2000.Q1- 2014.Q3	NA	1995.Q1- 2014.Q4
$Cons$	1996.Q1- 2014.Q4	2000.Q1- 2014.Q3	1990.Q1- 2014.Q4	1991.Q1- 2014.Q3	1990.Q1- 2014.Q3	1995.Q1- 2014.Q3	1993.Q1- 2014.Q3	2005.Q1- 2014.Q4	2001.Q1- 2014.Q4	1993.Q1- 2014.Q4	2000.Q1- 2014.Q3	NA	2000.Q1- 2014.Q4
Inv	1990.Q1- 2014.Q4	2000.Q1- 2014.Q4	1990.Q1- 2014.Q4	1991.Q1- 2014.Q4	1990.Q1- 2014.Q4	1995.Q1- 2014.Q4	2004.Q1- 2014.Q4	2005.Q1- 2014.Q4	2001.Q1- 2014.Q4	1993.Q1- 2014.Q4	1990.Q1- 2014.Q4	1998.Q1- 2014.Q3	2000.Q1- 2014.Q4
TB/Y	1990.Q1- 2014.Q4	1994.Q1- 2014.Q3	1991.Q1- 2014.Q4	1994.Q1- 2014.Q3	1990.Q1- 2014.Q4	1993.Q3- 2014.Q4	2004.Q1- 2014.Q4	1991.Q1- 2014.Q4	2001.Q1- 2014.Q4	1993.Q1- 2014.Q4	2000.Q1- 2014.Q3	NA	1995.Q1- 2014.Q4
$EMBI$	1999.Q2- 2015.Q1	1997.Q1- 2015.Q1	1997.Q2- 2015.Q1	1994.Q3- 2015.Q1	1995.Q1- 2015.Q1	1997.Q4- 2015.Q1	1994.Q1- 2015.Q1	1996.Q4- 2015.Q1	2000.Q2- 2015.Q1	1994.Q1- 2015.Q1	1995.Q2- 2015.Q1	1994.Q1- 2015.Q1	1995.Q2- 2013.Q4
$CEMBI$	2009.Q3- 2014.Q4	2003.Q4- 2014.Q4	2005.Q3- 2014.Q4	2002.Q1- 2014.Q4	2009.Q4- 2014.Q4	2002.Q2- 2014.Q4	2014.Q1- 2014.Q4	2002.Q1- 2014.Q4	2006.Q3- 2014.Q4	2002.Q1- 2014.Q4	NA	2002.Q1- 2013.Q2	NA
RER	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1991.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1993.Q4- 2014.Q4	1991.Q1- 2014.Q4	1980.Q1- 2014.Q4	1992.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1980.Q1- 2014.Q4	1992.Q1- 2014.Q4
RG	1990.Q1- 2014.Q4	2000.Q1- 2014.Q4	1990.Q1- 2014.Q4	1991.Q1- 2014.Q4	1990.Q1- 2014.Q4	1995.Q1- 2014.Q4	2004.Q1- 2014.Q4	1996.Q1- 2014.Q4	2001.Q1- 2014.Q4	1990.Q1- 2014.Q4	1990.Q1- 2014.Q4	NA	1997.Q1- 2014.Q4

Note: This table shows quarterly data availability for each variable (rows) for the 13 countries that compose EME13 (columns). P^{Co} was constructed using data from the International Monetary Fund's (IMF) monthly series on commodity prices and the United Nations' Comtrade dataset; further discussion on the index construction may be found in the core paper. The following variables: RGDP, GDP/CPI, GDP-G, GDP-G/CPI, Cons, Inv, TB/Y, RG come from HAVER Analytics databases. RER comes from IMF's IFS. CEMBI and EMBI come from Bloomberg.

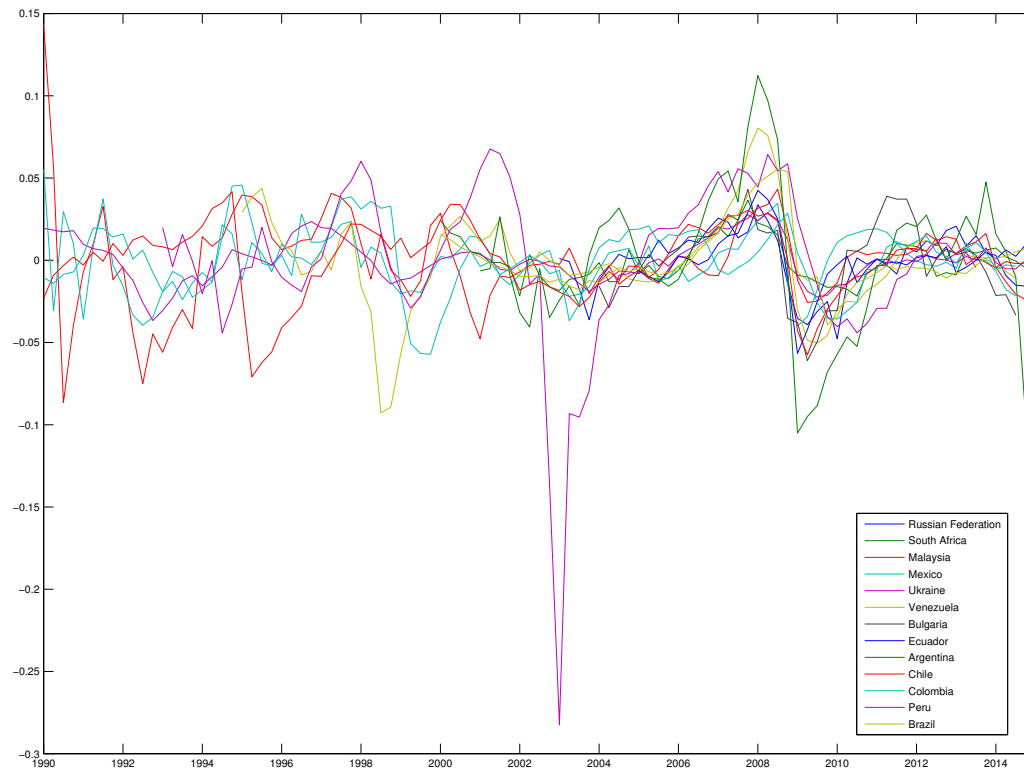
d. Comovement

Figure A3. Country-specific Commodity Price Indices in Emerging Economies: 1990-2014



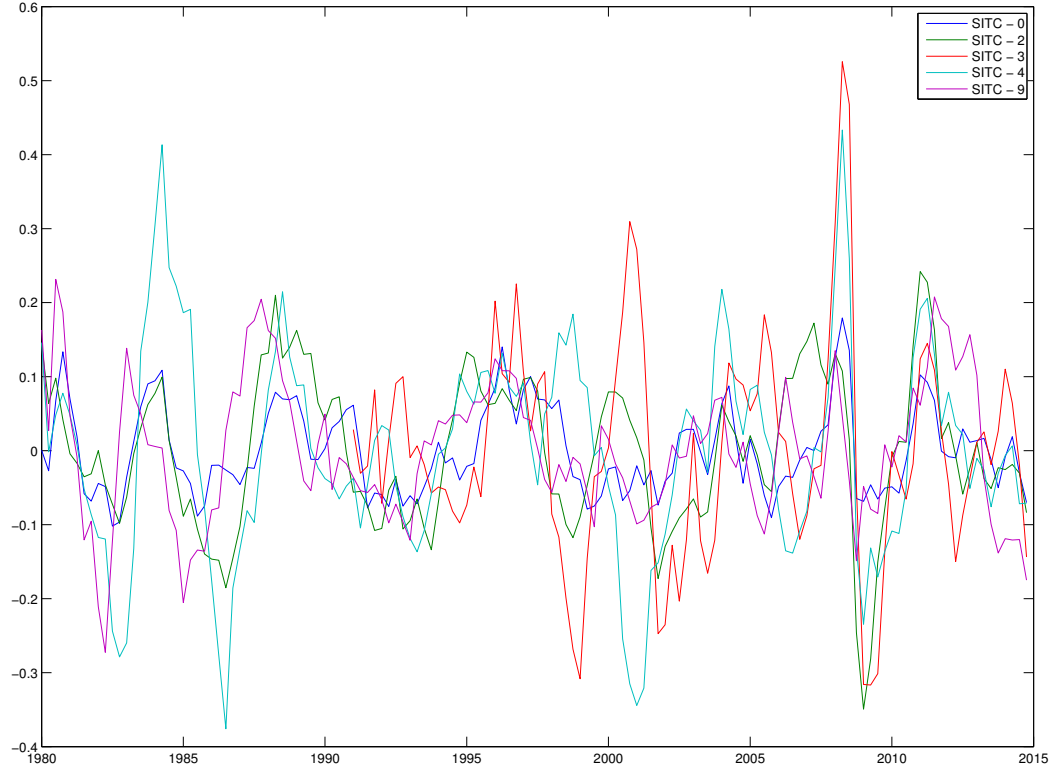
Note: This is the colored version of Figure 2 in the main text except that it includes now a legend. The figure reports the quarterly time series for the cyclical components of country-specific commodity price indices across the 13 EMEs studied (see footnote 7 in main text). Cyclical components are computed as relative deviations from a Hodrick-Prescott trend (λ of 1600). See Section 2 and this Appendix for the sources, and description of the construction of the indices.

Figure A4. Country-specific Real GDP: Time Series Plots



Note: Real GDP comes from Haver Analytics. Data are filtered using the Hodrick-Prescott filter ($\lambda=1600$).

Figure A5. Commodity Price Indices by SITC: Time Series Plots



Note: SITC Classification: 1 (beef, lamb, pork, chicken), 3 (fish, shrimp), 4 (wheat, rice, barley, corn, groundnuts, soybeans), 5 (bananas), 6 (sugar), 7 (coffee, cocoa, tea), 8 (soybean meal, fish meal), 21 (hides), 23 (natural rubber), 26 (hard log, soft log, hard sawn, soft sawn, cotton, wool), 28 (iron, copper, nickel, aluminum, lead, zinc, tin), 32 (coal), 33 (crude oil), 34 (natural gas), 42 (soy oil, olive oil, sun oil, palm oil, coconut oil), 97 (gold). Variables are detrended using the Hodrick-Prescott filter (lambda=1600).

Table A7. Principal Component Analysis: PComs & RGDP & SITC: LAC4 & EME13

Share of the variance explained by the first j components in:			
EME13			SITC
First j Principal Components	PCo	$RGDP$	$SITC - 1 \text{ digit}$
$j = 1$	0.78	0.76	0.55
$j = 2$	0.87	0.88	0.81
$j = 3$	0.92	0.93	0.93
LAC4			
First j Principal Components	PCo	$RGDP$	
$j = 1$	0.81	0.67	
$j = 2$	0.93	0.83	
$j = 3$	0.98	0.92	

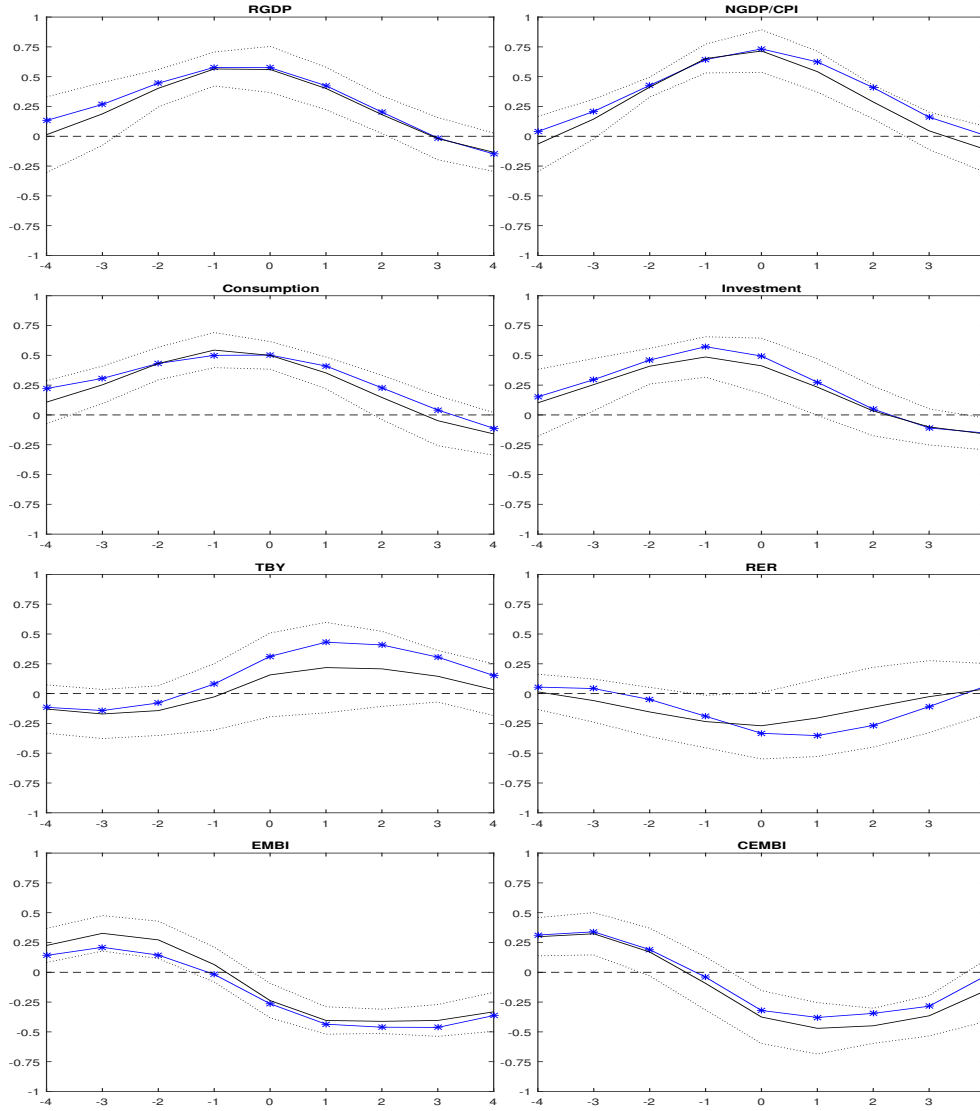
Note: This table shows the principal components analysis for $j = 1, 2, 3$ for two income groups. Variables are detrended using the Hodrick-Prescott filter (lambda=1600). Numbers are in percentages.

e. Other Filters

Sections (i)-(iii) present the same descriptive statistics presented above using three alternative detrending methods different from the HP-filter used in the baseline results: (i) growth rates; (ii) log-linear detrending; (iii) log-quadratic detrending.

i. Growth Rates

Figure A6. Unconditional Serial Correlations with Commodity Price Indices



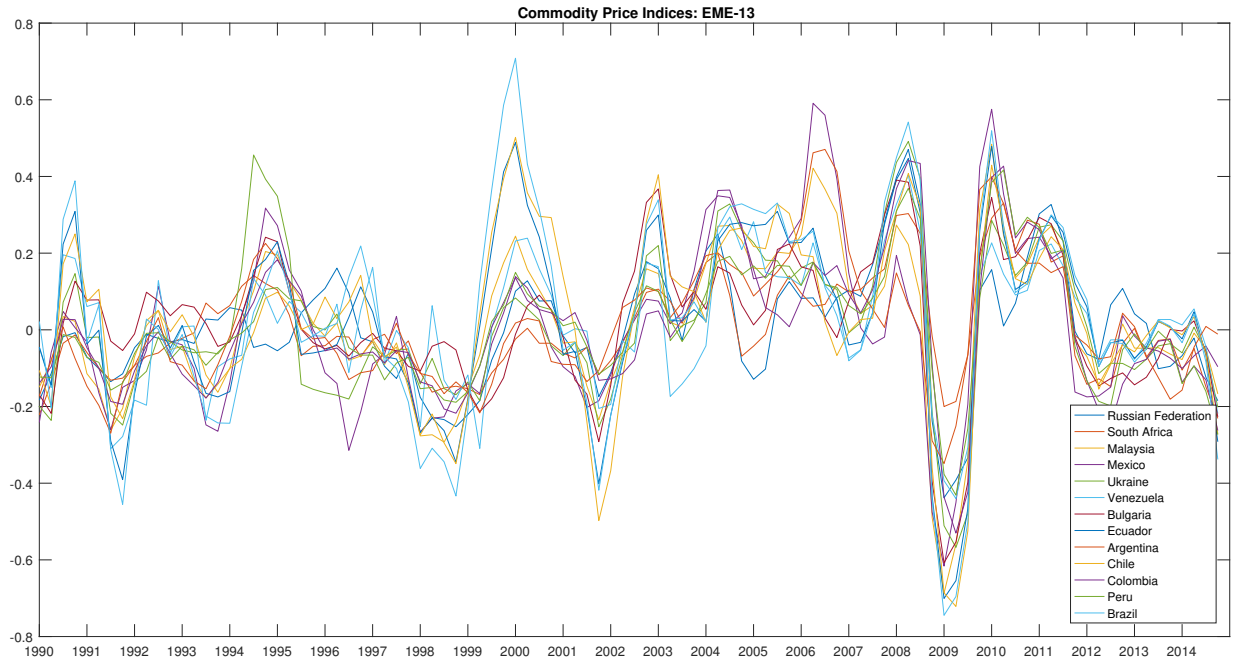
Note: Data is filtered using growth rates, instead of the Hodrick-Prescott filter. Each panel reports in black/solid lines the average correlation between the macroeconomic variable in the subtitle in period t and the country-specific commodity price indices in $t+j$ with $j = -4, \dots, 4$, across 13 EMEs. Dashed/gray lines report plus/minus one S.D. bands. Blue/solid line with “*” report the results for a smaller sub-group of EMEs in Latin America (Brazil, Chile, Colombia, and Peru). All statistics are computed using growth rates (instead of using the Hodrick-Prescott Filter) on a quarterly panel between 2000.Q1 and 2014.Q4.

Table A8. Second Moments: All EMEs-13 & LAC4

	Latin America and the Caribbean (LAC 4)												
	S.D. (%)	Relative SD to RGDP	Autocorrelations		Cross-correlation with $RGDP_{t+j}$								
			1	2	$j = -4$	$j = -3$	$j = -2$	$j = -1$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$
PCo	19.02	7.82	0.79	0.42	0.13	0.27	0.45	0.58	0.58	0.57	0.46	0.31	0.19
$RGDP$	2.53	1.00	0.78	0.48	-0.10	0.17	0.48	0.78	1.00	0.78	0.48	0.17	-0.10
GDP/CPI	4.17	1.64	0.76	0.50	0.03	0.22	0.45	0.64	0.77	0.72	0.57	0.38	0.20
$RGDP-G$	3.07	1.21	0.78	0.48	-0.12	0.16	0.47	0.76	0.97	0.77	0.48	0.19	-0.08
$GDP-G/CPI$	4.70	1.86	0.76	0.49	-0.03	0.21	0.44	0.65	0.77	0.72	0.56	0.36	0.18
$Cons$	2.21	0.88	0.84	0.65	0.08	0.31	0.50	0.70	0.81	0.71	0.51	0.27	0.04
Inv	8.67	3.41	0.74	0.43	-0.12	0.08	0.32	0.61	0.80	0.63	0.32	0.07	-0.15
TB/GDP	2.75	1.06	0.68	0.40	-0.03	-0.02	-0.02	-0.05	-0.05	0.05	0.15	0.23	0.25
$EMBIG$	42.91	17.40	0.66	0.31	-0.20	-0.15	-0.08	-0.04	-0.05	-0.13	-0.27	-0.40	-0.43
$CEMBI$	35.10	15.09	0.64	0.21	0.30	0.42	0.37	0.27	0.13	0.01	-0.09	-0.16	-0.12
RER	8.12	3.39	0.72	0.34	-0.05	-0.18	-0.31	-0.36	-0.31	-0.25	-0.18	-0.12	-0.08
Emerging Market Economies (EME 13)													
PCo	21.00	6.88	0.76	0.32	0.01	0.19	0.40	0.56	0.56	0.53	0.35	0.15	0.00
$RGDP$	3.67	1.00	0.81	0.54	-0.02	0.25	0.54	0.81	1.00	0.81	0.54	0.25	-0.02
GDP/CPI	5.30	1.65	0.78	0.50	0.00	0.18	0.42	0.64	0.79	0.70	0.50	0.27	0.05
$RGDP-G$	4.45	1.36	0.81	0.51	-0.08	0.18	0.49	0.77	0.95	0.80	0.53	0.25	-0.01
$GDP-G/CPI$	6.53	2.02	0.76	0.48	-0.04	0.15	0.39	0.64	0.79	0.71	0.50	0.27	0.07
$Cons$	3.34	0.98	0.82	0.61	0.11	0.32	0.52	0.71	0.80	0.69	0.48	0.26	0.03
Inv	11.25	3.09	0.81	0.55	0.10	0.32	0.53	0.72	0.78	0.67	0.44	0.20	-0.01
TB/GDP	3.53	1.13	0.56	0.32	-0.09	-0.13	-0.18	-0.17	-0.14	-0.09	-0.03	0.04	0.06
$EMBIG$	54.51	17.15	0.70	0.36	-0.09	-0.11	-0.09	-0.10	-0.13	-0.19	-0.24	-0.29	-0.30
$CEMBI$	46.11	14.60	0.73	0.35	0.33	0.30	0.16	0.00	-0.16	-0.25	-0.28	-0.28	-0.22
RER	10.48	3.31	0.77	0.46	0.00	-0.03	-0.12	-0.22	-0.27	-0.24	-0.17	-0.11	-0.09

Note: This table shows the standard deviation (first column), standard deviation relative to RGDP (second column), first and second autocorrelations (third and fourth column). and the cross-correlation with RGDP in $t = -4, \dots, 4$ for two country groups. LAC4 is composed by: Brazil, Colombia, Chile and Peru. EME13 is composed by LAC4, Bulgaria, Ecuador, Venezuela, Malaysia, Ukraine, Mexico, Argentina, South Africa and the Russian Federation. Calculations were made with a sample from 2000.Q1 to 2014.Q4. Data (except for the trade balance) are expressed in natural logarithms before filtering using growth rates.

Figure A7. Commodity Price Indices: EME-13



Note: Data filtered with growth rates.

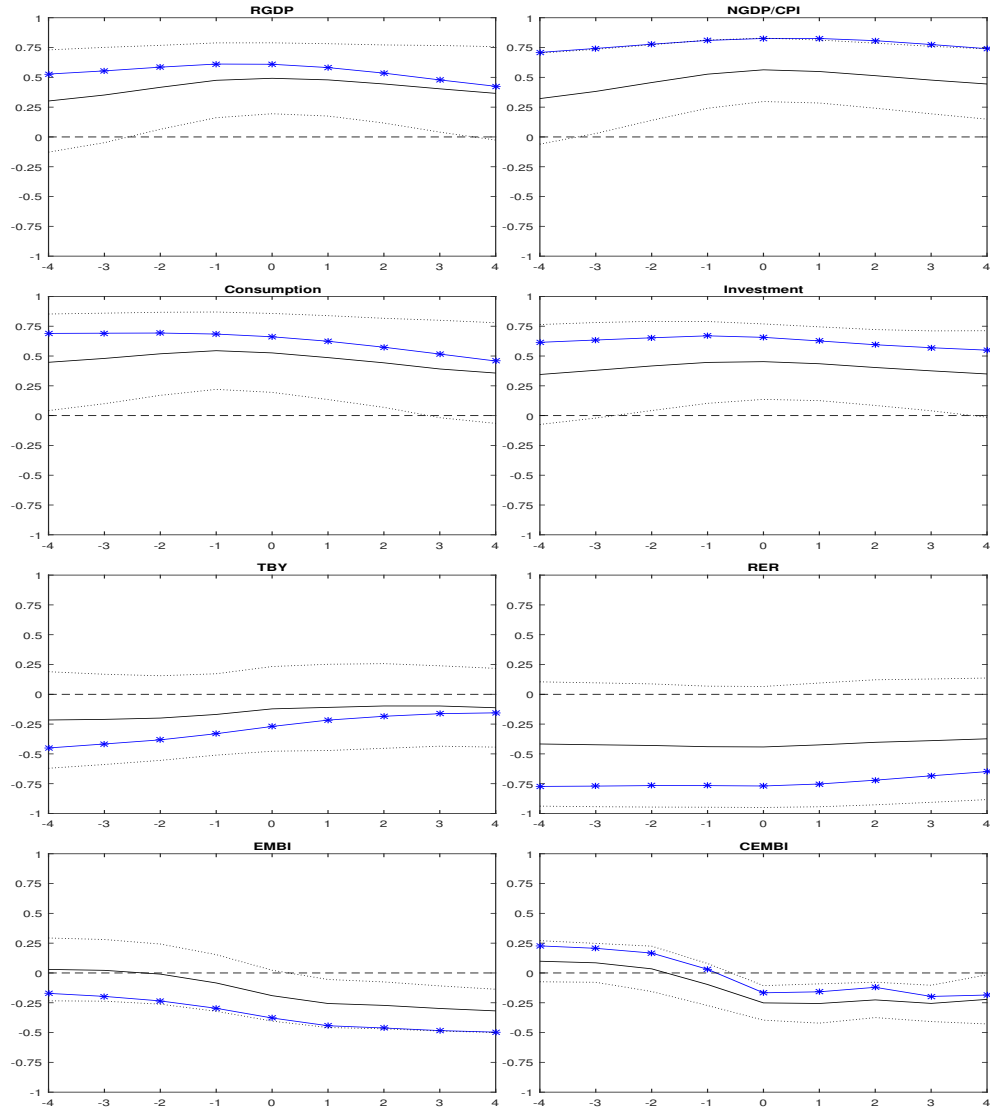
Table A9. Principal Component Analysis: PComs & RGDP & SITC: LAC4 & EME13

Share of the variance explained by the first j components in:			
EME13			SITC
First j Principal Components	PCo	$RGDP$	$SITC - 1\ digit$
$j = 1$	0.80	0.76	0.59
$j = 2$	0.88	0.88	0.82
$j = 3$	0.93	0.91	0.93
LAC4			
First j Principal Components	PCo	$RGDP$	
$j = 1$	0.83	0.68	
$j = 2$	0.94	0.82	
$j = 3$	0.99	0.92	

Note: This table shows the principal components analysis for $j = 1, 2, 3$ for two income groups. Variables are detrended using growth rates. Numbers are in percentages.

ii. Log-linear Detrending

Figure A8. Unconditional Serial Correlations with Commodity Price Indices



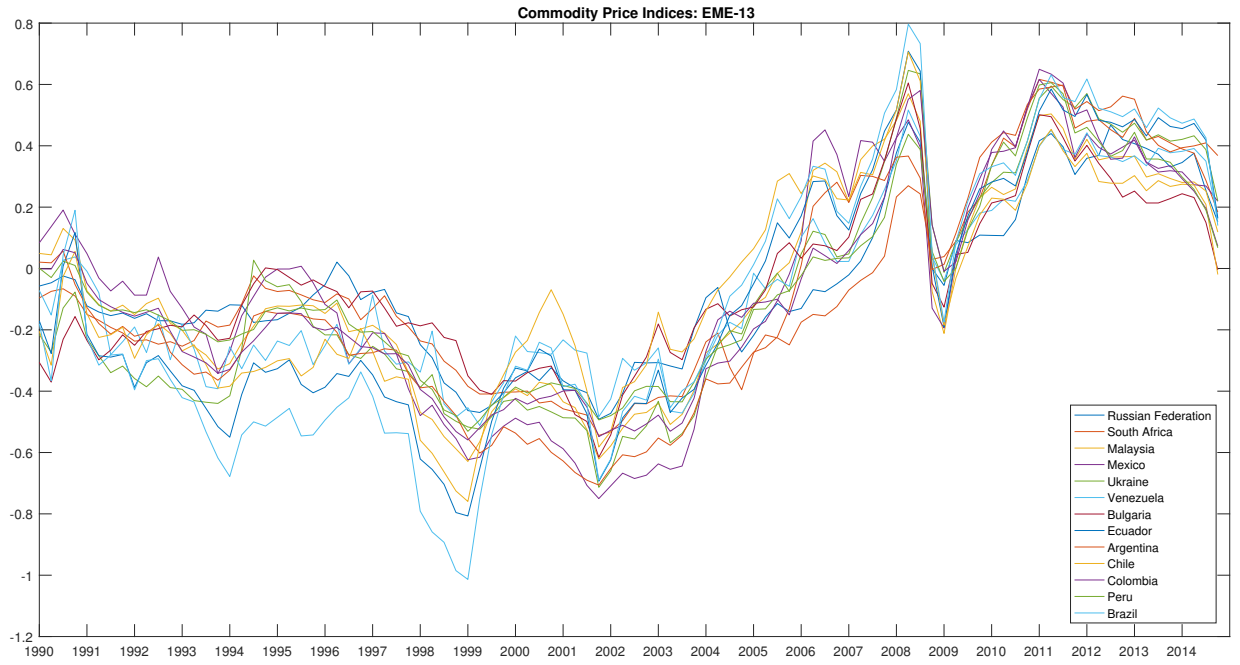
Note: Data is filtered using log-linear detrending, instead of the Hodrick-Prescott filter. Each panel reports in black/solid lines the average correlation between the macroeconomic variable in the subtitle in period t and the country-specific commodity price indices in $t+j$ with $j = -4, \dots, 4$, across 13 EMEs. Dashed/gray lines report plus/minus one S.D. bands. Blue/solid line with “*” report the results for a smaller sub-group of EMEs in Latin America (Brazil, Chile, Colombia, and Peru). All statistics are computed using growth rates (instead of using the Hodrick-Prescott Filter) on a quarterly panel between 2000.Q1 and 2014.Q4.

Table A10. Second Moments: All EMEs-13 & LAC4

	Latin America and the Caribbean (LAC 4)												
	S.D. (%)	Relative SD to RGDP	Autocorrelations		Cross-correlation with $RGDP_{t+j}$								
			1	2	$j = -4$	$j = -3$	$j = -2$	$j = -1$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$
PCo	42.20	15.90	0.98	0.94	0.53	0.55	0.59	0.61	0.61	0.64	0.63	0.60	0.56
$RGDP$	3.13	1.00	0.91	0.80	0.55	0.67	0.80	0.91	1.00	0.91	0.80	0.67	0.55
GDP/CPI	5.93	2.25	0.94	0.86	0.56	0.62	0.69	0.72	0.74	0.76	0.74	0.69	0.62
$RGDP-G$	3.53	1.15	0.90	0.78	0.44	0.58	0.72	0.85	0.96	0.89	0.80	0.69	0.58
$GDP-G/CPI$	5.47	1.88	0.88	0.76	0.49	0.62	0.72	0.80	0.85	0.85	0.81	0.73	0.64
$Cons$	4.14	1.33	0.97	0.92	0.63	0.70	0.76	0.80	0.81	0.75	0.65	0.54	0.41
Inv	11.67	3.71	0.90	0.81	0.47	0.54	0.62	0.71	0.78	0.73	0.64	0.57	0.49
TB/GDP	3.26	1.17	0.86	0.78	-0.19	-0.15	-0.14	-0.15	-0.13	-0.09	-0.04	0.01	0.04
$EMBIG$	38.96	14.80	0.81	0.61	-0.35	-0.32	-0.29	-0.26	-0.22	-0.31	-0.39	-0.46	-0.52
$CEMBI$	26.26	10.29	0.67	0.27	0.17	0.21	0.18	0.18	0.14	0.07	0.00	-0.10	-0.10
RER	12.89	5.33	0.94	0.86	-0.47	-0.50	-0.53	-0.54	-0.52	-0.51	-0.48	-0.43	-0.37
Emerging Market Economies (EME 13)													
PCo	36.44	10.78	0.96	0.89	0.30	0.35	0.42	0.48	0.49	0.49	0.44	0.39	0.33
$RGDP$	4.57	1.00	0.93	0.82	0.56	0.69	0.82	0.93	1.00	0.93	0.82	0.69	0.56
GDP/CPI	6.14	1.72	0.91	0.79	0.39	0.48	0.58	0.66	0.70	0.68	0.61	0.51	0.41
$RGDP-G$	4.81	1.27	0.91	0.76	0.41	0.55	0.69	0.82	0.90	0.84	0.74	0.62	0.50
$GDP-G/CPI$	6.85	1.87	0.88	0.75	0.36	0.47	0.59	0.69	0.75	0.73	0.65	0.56	0.45
$Cons$	5.01	1.30	0.91	0.81	0.53	0.61	0.66	0.71	0.73	0.66	0.55	0.44	0.33
Inv	14.15	3.17	0.93	0.83	0.57	0.66	0.74	0.81	0.84	0.81	0.74	0.65	0.54
TB/GDP	3.97	1.18	0.78	0.68	-0.21	-0.21	-0.23	-0.23	-0.21	-0.19	-0.16	-0.12	-0.11
$EMBIG$	53.99	14.26	0.85	0.67	-0.27	-0.33	-0.34	-0.35	-0.34	-0.38	-0.40	-0.41	-0.40
$CEMBI$	37.40	10.90	0.76	0.45	0.15	0.10	0.00	-0.07	-0.14	-0.19	-0.23	-0.27	-0.25
RER	13.13	3.81	0.93	0.83	-0.31	-0.33	-0.39	-0.43	-0.45	-0.46	-0.45	-0.41	-0.38

Note: This table shows the standard deviation (first column), standard deviation relative to RGDP (second column), first and second autocorrelations (third and fourth column). and the cross-correlation with RGDP in $t = -4, \dots, 4$ for two country groups. LAC4 is composed by: Brazil, Colombia, Chile and Peru. EME13 is composed by LAC4, Bulgaria, Ecuador, Venezuela, Malaysia, Ukraine, Mexico, Argentina, South Africa and the Russian Federation. Calculations were made with a sample from 2000.Q1 to 2014.Q4. Data (except for the trade balance) are expressed in natural logarithms before filtering using log-linear detrending.

Figure A9. Commodity Price Indices: EME-13



Note: Data filtered with log-linear detrending.

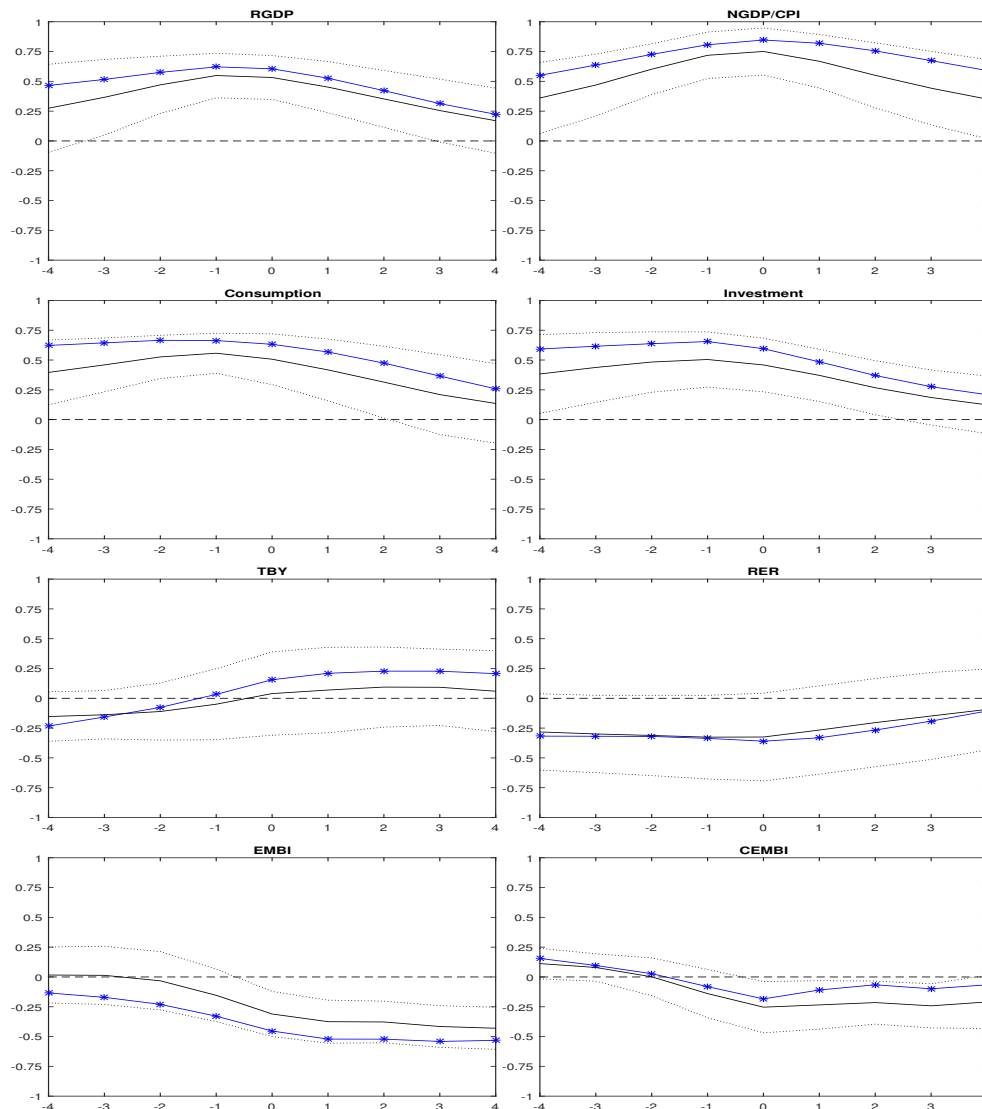
Table A11. Principal Component Analysis: PComs & RGDP & SITC: LAC4 & EME13

Share of the variance explained by the first j components in:			
EME13			SITC
First j Principal Components	PC_o	$RGDP$	$SITC - 1 \text{ digit}$
$j = 1$	0.92	0.67	0.82
$j = 2$	0.96	0.86	0.93
$j = 3$	0.98	0.93	0.98
LAC4			
First j Principal Components	PC_o	$RGDP$	
$j = 1$	0.95	0.79	
$j = 2$	0.98	0.93	
$j = 3$	1.00	0.98	

Note: This table shows the principal components analysis for $j = 1, 2, 3$ for two income groups. Variables are detrended using log-linear detrending. Numbers are in percentages.

iii. Log-quadratic Detrending

Figure A10. Unconditional Serial Correlations with Commodity Price Indices



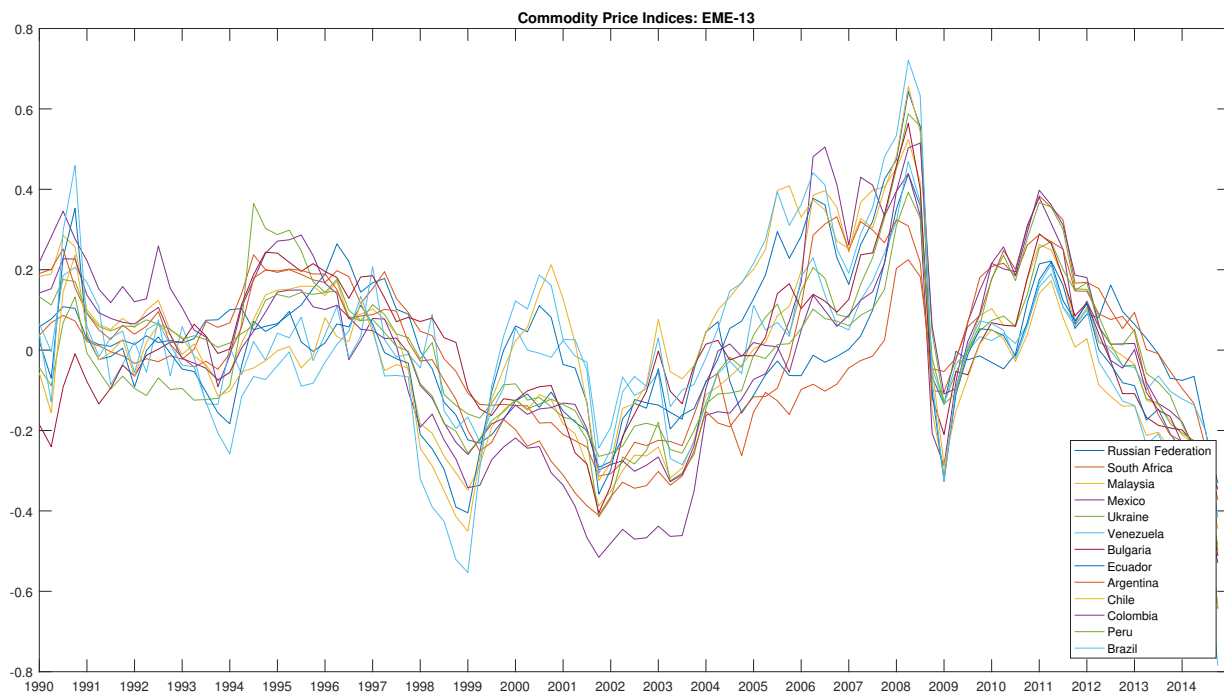
Note: Data is filtered using log-quadratic detrending, instead of the Hodrick-Prescott filter. Each panel reports in black/solid lines the average correlation between the macroeconomic variable in the subtitle in period t and the country-specific commodity price indices in $t+j$ with $j = -4, \dots, 4$, across 13 EMEs. Dashed/gray lines report plus/minus one S.D. bands. Blue/solid line with “*” report the results for a smaller sub-group of EMEs in Latin America (Brazil, Chile, Colombia, and Peru). All statistics are computed using growth rates (instead of using the Hodrick-Prescott Filter) on a quarterly panel between 2000.Q1 and 2014.Q4.

Table A12. Second Moments: All EMEs-13 & LAC4

	Latin America and the Caribbean (LAC 4)												
	S.D. (%)	Relative SD to RGDP	Autocorrelations		Cross-correlation with $RGDP_{t+j}$								
			1	2	$j = -4$	$j = -3$	$j = -2$	$j = -1$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$
P^{Co}	23.80	10.56	0.93	0.81	0.47	0.52	0.58	0.62	0.61	0.65	0.63	0.59	0.53
$RGDP$	2.36	1.00	0.88	0.73	0.43	0.58	0.73	0.88	1.00	0.88	0.73	0.58	0.43
GDP/CPI	4.66	2.04	0.90	0.79	0.40	0.50	0.62	0.69	0.74	0.77	0.73	0.66	0.57
$RGDP-G$	2.84	1.20	0.88	0.73	0.34	0.50	0.68	0.85	0.97	0.88	0.75	0.61	0.47
$GDP-G/CPI$	4.91	2.11	0.87	0.73	0.33	0.48	0.61	0.72	0.80	0.80	0.74	0.64	0.53
$Cons$	2.56	1.09	0.94	0.85	0.55	0.64	0.71	0.77	0.78	0.72	0.61	0.46	0.30
Inv	9.02	3.72	0.85	0.71	0.40	0.49	0.59	0.70	0.77	0.70	0.56	0.44	0.33
TB/GDP	3.26	1.39	0.86	0.78	-0.19	-0.14	-0.10	-0.07	-0.02	0.02	0.07	0.12	0.15
$EMBIG$	38.24	16.95	0.80	0.60	-0.33	-0.32	-0.30	-0.29	-0.28	-0.37	-0.45	-0.52	-0.56
$CEMBI$	22.63	11.09	0.58	0.11	0.24	0.28	0.23	0.20	0.09	-0.01	-0.09	-0.16	-0.12
RER	9.09	4.06	0.86	0.69	-0.38	-0.43	-0.46	-0.46	-0.39	-0.39	-0.36	-0.31	-0.24
Emerging Market Economies (EME 13)													
P^{Co}	22.25	8.01	0.89	0.71	0.27	0.37	0.47	0.55	0.53	0.54	0.48	0.39	0.30
$RGDP$	3.34	1.00	0.89	0.73	0.39	0.56	0.73	0.89	1.00	0.89	0.73	0.56	0.39
GDP/CPI	5.35	1.87	0.89	0.74	0.31	0.43	0.56	0.67	0.74	0.69	0.59	0.46	0.32
$RGDP-G$	3.60	1.28	0.87	0.68	0.27	0.44	0.63	0.81	0.93	0.84	0.69	0.52	0.36
$GDP-G/CPI$	6.11	2.16	0.86	0.69	0.27	0.40	0.55	0.68	0.76	0.72	0.60	0.47	0.33
$Cons$	3.54	1.22	0.88	0.75	0.42	0.53	0.63	0.72	0.75	0.67	0.55	0.41	0.26
Inv	10.95	3.30	0.88	0.73	0.44	0.57	0.68	0.77	0.81	0.76	0.63	0.48	0.34
TB/GDP	3.97	1.54	0.78	0.68	-0.15	-0.16	-0.18	-0.17	-0.13	-0.10	-0.07	-0.02	0.00
$EMBIG$	48.08	17.04	0.82	0.61	-0.18	-0.24	-0.25	-0.26	-0.26	-0.32	-0.35	-0.37	-0.38
$CEMBI$	34.56	11.93	0.72	0.37	0.22	0.18	0.08	-0.01	-0.12	-0.21	-0.26	-0.30	-0.27
RER	11.02	3.76	0.88	0.72	-0.16	-0.19	-0.25	-0.28	-0.29	-0.27	-0.23	-0.18	-0.13

Note: This table shows the standard deviation (first column), standard deviation relative to RGDP (second column), first and second autocorrelations (third and fourth column). and the cross-correlation with RGDP in $t = -4, \dots, 4$ for two country groups. LAC4 is composed by: Brazil, Colombia, Chile and Peru. EME13 is composed by LAC4, Bulgaria, Ecuador, Venezuela, Malaysia, Ukraine, Mexico, Argentina, South Africa and the Russian Federation. Calculations were made with a sample from 2000.Q1 to 2014.Q4. Data (except for the trade balance) are expressed in natural logarithms before filtering using log-quadratic detrending.

Figure A11. Commodity Price Indices: EME-13



Note: Data filtered with log-quadratic detrending.

Table A13. Principal Component Analysis: PComs & RGDP & SITC: LAC4 & EME13

Share of the variance explained by the first j components in:			
EME13			SITC
First j Principal Components	P^{Co}	$RGDP$	$SITC - 1 \text{ digit}$
$j = 1$	0.78	0.65	0.67
$j = 2$	0.91	0.87	0.88
$j = 3$	0.95	0.94	0.96
LAC4			
First j Principal Components	P^{Co}	$RGDP$	
$j = 1$	0.88	0.66	
$j = 2$	0.95	0.88	
$j = 3$	0.99	0.95	

Note: This table shows the principal components analysis for $j = 1, 2, 3$ for two income groups. Variables are detrended using log-quadratic detrending. Numbers are in percentages.

f. Commodity Export Shares

An empirical task that we set out to do but that we ended up not reporting in the main text is to assess the size of the share of commodity exports in a typical EME. To answer this we assemble a large (unbalanced) annual panel using the World Bank's World Development Indicators (WDI) covering 189 countries between the years 1960 and 2013. For each

country, the panel contains the share of three broad groups of commodities in total exports: agricultural, fuel and metals. We average the sum of these three shares across time to obtain country-specific commodity export shares. Finally we separate countries between 61 EMEs and 128 non-EMEs. The latter is also further subcategorized into 74 advanced and 54 low-income economies¹.

The main descriptive statistics are reported in Tables A14. The table documents the median export share across all groups of countries.² It also displays Mann-Whitney tests for the equivalence between non-EMEs groups and EMEs, and Kruskal-Wallis for equivalence across all groups. The results in this Table allow us to define a first stylized fact: **The share of commodities in total exports in the average EME is more than double that of advanced economies.** While the median export share in EMEs is 25.7 percent, that in advanced is 11.2 percent. Furthermore, a Mann - Whitney test easily rejects equivalence between the two distributions at 1 percent significance level. It fails to reject that hypothesis when emerging economies are compared to low-income economies, but continues to accept it when EMEs are compared to non-EMEs, mainly due to the smaller number of observations for low-income countries relative to those for advanced economies. Kruskal-Wallis tests easily reject equivalence across these three groups of countries. This stylized fact is robust to two alternative measures of the relative importance of commodities: the gross and net share of commodity exports to GDP (see Table A.15).

Table A14. Commodity Export Shares: Benchmark

	All Countries	Advanced	Emerging	Low Income	Non-Emerging
Median (%)	19.3	11.2	25.7	26.5	14
Mann-Whitney Test (against Emerging)		3.046 *** (0.0023)		-0.689 (0.4906)	-2.309 ** (0.0209)
Kruskal-Wallis Test		8.048 ** (0.0179)			
No. of Countries	189	74	61	54	128
No. of observations (annual)	5514	2296	2205	1013	3309

Note: The first row reports the median commodity export share in total exports across various groups of countries (in percentages). Second and third rows report Mann-Whitney and Kruskal – Wallis tests statistics (p-values in parenthesis). Mann-Whitney tests are computed always relative to emerging economies. The Kruskal – Wallis test reports the results using three groups: advanced, emerging and low income. Data come from WDI in the form of an unbalanced annual panel of three kinds of commodity exports (agricultural, fuels, and metals). EMEs are classified as such if data on EMBI exist. Advanced and low-income are categorized according to World Bank’s WDI indicators. The Appendix contains further information on the countries in each group and a description of the dataset. Statistical significance at 1, 5, and 10 percent is reported with (***), (**), and (*), respectively.

¹The 61 EMEs in our sample are classified as such following a simple criterion: we classify a country as an EME if there exists EMBI data on this country. The only two countries that we exclude from this list are China and India, as they clearly do not fall into the category of *small and open* emerging economy that is the focus of our analysis. Advanced and low-income economies are classified following World Bank’s classification.

²We report medians instead of means as the distributions in each of the groups are highly skewed and simple tests reject normality.

Table A15. Commodity Export Shares: Across country-specific averages; across observations; net commodity exports; commodities export

	All countries	Advanced	Emerging	Low Income	Non-Emerging
	<i>Across country-specific averages</i>				
Median	19.3	11.2	25.7	26.5	14
Mann-Whitney Test (against emerging)		3.046 *** (0.0023)		-0.689 (0.4906)	-2.309** (0.0209)
Kruskal-Wallis Test		8.048 ** (0.0179)			5.333 ** (0.0209)
No. of Countries	189	74	61	54	128
	<i>Across observations</i>				
Median	18.5	10.7	23.2	30.4	13.9
Mann-Whitney Test (against emerging)		17.544*** (0.0000)		3.426 *** (0.0006)	-11.755*** (0.0000)
Kruskal-Wallis Test		364.329*** (0.0001)			138.18*** (0.0001)
No. of Observations	5514	2296	2205	1013	3309
	<i>Net commodities (as % of current GDP)</i>				
Median	-1.8256	-2.5640	-1.7963	-0.858	-1.8737
Mann-Whitney Test (against emerging)		1.039 (0.2989)		0.323 (0.7468)	-0.514 (0.6073)
Kruskal-Wallis Test		1.643 (0.4398)			
No. of Countries	178	69	60	49	118
No. of observations (annual)	5108	2127	2046	935	3062
	<i>Commodities exports (as % of current GDP)</i>				
Median	3.6146	1.931	4.6068	5.6594	2.4181
Mann-Whitney Test (against emerging)		2.522 ** (0.0117)		-1.511 (0.1309)	-2.406** (0.0161)
Kruskal-Wallis Test		6.111 ** (0.0471)			
No. of Countries	178	69	60	49	118
No. of observations (annual)	5108	2127	2046	935	3062

Note: The first row reports the median commodity export share in total exports across various groups of countries (in percentages). Second and third rows report Mann-Whitney and Kruskal – Wallis tests statistics (p-values in parenthesis). Mann-Whitney tests are computed always relative to emerging economies. The Kruskal – Wallis test reports the results using three groups: advanced, emerging and low income. Data come from WDI in the form of an unbalanced annual panel of three kinds of commodity exports (agricultural, fuels, and metals). EMEs are classified as such if data on EMBI exist. Advanced and low-income are categorized according to World Bank's WDI indicators. The Appendix contains further information on the countries in each group and a description of the dataset. Statistical significance at 1, 5, and 10 percent is reported with (***), (**), and (*), respectively.

Table A16. Commodity Export Shares per country: Low Income, EMEs, and Developed

Low Income Countries: 54 countries							
Country	Obs	Coverage	Median	Country	Obs	Coverage	Median
Afghanistan	3	1967:1976	50.84	Rwanda	15	1996:2012	38.64
Bangladesh	21	1977:2007	3.05	Samoa	20	1973:2012	1.26
Benin	34	1962:2010	35.55	Sao Tome and Principe	6	2003:2012	0.61
Bhutan	13	1991:2011	38.24	Sierra Leone	7	1972:1984	18.02
Burkina Faso	37	1962:2011	45.58	Solomon Islands	3	2010:2012	32.79
Burundi	12	1965:2012	12.94	Somalia	14	1962:1982	8.43
Cabo Verde	13	1978:2011	0.31	Sudan	36	1963:2011	69.17
Cambodia	20	1962:2012	10.52	Swaziland	8	2000:2007	9.65
Cameroon	40	1962:2012	40.67	Syrian Arab Republic	33	1974:2010	75.24
Central African Republic	25	1962:2011	48.25	Tajikistan	1	2000	82.26
Chad	14	1962:1975	80.77	Timor-Leste	2	2004:2005	0.28
Comoros	4	1962:2009	0.67	Togo	44	1962:2012	49.43
Congo, Dem. Rep.	10	1962:1978	75.76	Uganda	20	1976:2012	12.85
Congo, Rep.	29	1962:2010	74.00	Vanuatu	6	2000:2011	2.71
Djibouti	7	1986:2009	6.02	Yemen, Rep.	19	1991:2012	94.04
Eritrea	2	2001:2003	25.45	Zambia	30	1966:2011	87.80
Ethiopia	13	1993:2012	14.66	Zimbabwe	24	1984:2012	28.62
Gambia, The	17	1964:2011	4.35				
Guinea	13	1995:2008	72.53				
Guinea-Bissau	3	1971:2004	7.04				
Guyana	27	1970:2012	27.54				
Haiti	3	1990:1996	0.70				
Kenya	34	1976:2010	21.96				
Kiribati	3	2005:2008	3.01				
Kyrgyz Republic	17	1995:2012	33.88				
Lesotho	6	2001:2009	3.28				
Liberia	16	1967:1984	91.94				
Madagascar	47	1962:2012	13.29				
Malawi	43	1966:2011	3.04				
Mali	30	1962:2012	50.45				
Mauritania	17	1962:2012	81.63				
Moldova	17	1994:2012	5.31				
Myanmar	18	1962:2010	34.63				
Nepal	8	1985:2011	5.59				
Nicaragua	45	1965:2012	15.09				
Niger	37	1962:2012	50.45				
Papua New Guinea	27	1971:2012	64.68				

Note: This table shows the number of available observations by income groups, EMEs are classified as such if data on EMBI exist. Advanced and low-income are categorized according to World Bank's WDI indicators. The "Median" column refers to the country-specific median of commodity exports shares, these are in percentages (%). Note that there might be gaps in the data coverage.

Table A16. Commodity Export Shares per country: Low Income, EMEs, and Developed (cont.)

Emerging Market Economies (EMEs): 61 countries							
Country	Obs	Coverage	Median	Country	Obs	Coverage	Median
Argentina	51	1962:2012	13.26	Lithuania	20	1992:2012	25.02
Armenia	15	1997:2012	31.83	Malaysia	49	1964:2012	40.33
Belarus	15	1998:2012	30.56	Mexico	51	1962:2012	28.65
Belize	26	1975:2012	4.44	Mongolia	12	1996:2007	75.96
Bolivia	51	1962:2012	87.64	Morocco	51	1962:2012	26.52
Brazil	50	1962:2012	17.99	Mozambique	18	1994:2012	69.26
Bulgaria	17	1996:2012	26.40	Namibia	13	2000:2012	27.15
Chile	51	1962:2012	69.69	Nigeria	40	1962:2012	94.35
Colombia	51	1962:2012	28.63	Pakistan	50	1962:2012	11.75
Costa Rica	48	1965:2012	4.38	Panama	50	1962:2011	10.11
Cote d'Ivoire	40	1962:2012	28.73	Paraguay	31	1967:2012	25.40
Croatia	21	1992:2012	18.53	Peru	50	1962:2012	60.49
Dominican Republic	31	1974:2012	2.89	Philippines	51	1962:2012	14.90
Ecuador	49	1962:2012	47.83	Poland	27	1984:2012	11.88
Egypt, Arab Rep.	48	1965:2012	54.11	Romania	24	1989:2012	14.88
El Salvador	50	1963:2012	6.53	Russian Federation	17	1996:2012	65.37
Gabon	34	1962:2009	95.11	Senegal	44	1962:2012	26.04
Georgia	17	1996:2012	28.24	Serbia	12	1992:2007	19.15
Ghana	39	1962:2012	25.23	Slovak Republic	19	1994:2012	10.38
Greece	51	1962:2012	20.77	South Africa	32	1974:2012	26.00
Guatemala	48	1965:2012	13.99	Sri Lanka	46	1962:2012	18.03
Honduras	46	1963:2012	11.74	Tanzania	22	1976:2012	26.92
Hungary	46	1966:2012	6.25	Thailand	50	1962:2012	13.23
Indonesia	47	1962:2012	56.02	Trinidad and Tobago	43	1968:2010	71.90
Iraq	14	1963:2009	91.61	Tunisia	50	1962:2011	26.84
Jamaica	43	1962:2012	20.69	Turkey	51	1962:2012	11.22
Jordan	47	1964:2012	32.42	Ukraine	17	1996:2012	16.18
Kazakhstan	18	1995:2012	78.27	Uruguay	42	1970:2012	17.96
Korea, Rep.	51	1962:2012	6.54	Venezuela, RB	49	1962:2011	95.51
Latvia	19	1994:2012	25.69	Vietnam	16	1997:2012	23.73
Lebanon	24	1967:2012	11.23				

Table A16. Commodity Export Shares per country: Low Income, EMEs, and Developed (cont.)

Advanced economies: 74 countries							
Country	Obs	Coverage	Median	Country	Obs	Coverage	Median
Albania	17	1996:2012	24.14	Japan	51	1962:2012	2.50
Algeria	45	1966:2012	97.03	Kuwait	38	1970:2009	92.59
Andorra	12	1995:2006	3.13	Libya	20	1962:2010	99.80
Angola	12	1962:1990	54.61	Luxembourg	14	1999:2012	7.01
Antigua and Barbuda	14	1973:2012	6.04	Macao SAR, China	32	1974:2010	1.09
Aruba	8	1988:2012	1.29	Macedonia, FYR	18	1994:2012	13.30
Australia	50	1963:2012	47.28	Maldives	5	2004:2011	1.57
Austria	50	1963:2012	9.70	Malta	47	1965:2012	1.45
Azerbaijan	17	1996:2012	88.55	Mauritius	32	1970:2012	0.75
Bahamas, The	28	1974:2012	22.73	Netherlands	51	1962:2012	17.21
Bahrain	39	1970:2011	80.67	New Caledonia	23	1975:2012	35.20
Barbados	46	1967:2012	1.86	New Zealand	49	1964:2012	27.49
Belgium	51	1962:2012	12.20	Norway	49	1962:2012	59.35
Bermuda	10	1971:2012	21.04	Oman	34	1979:2012	88.09
Bosnia and Herzegovina	10	2003:2012	31.79	Portugal	51	1962:2012	12.33
Botswana	13	2000:2012	11.53	Qatar	26	1972:2011	86.05
Brunei Darussalam	40	1962:2012	99.83	Saudi Arabia	34	1968:2012	91.56
Canada	51	1962:2012	34.30	Seychelles	28	1976:2008	0.31
Cayman Islands	1	1980	54.74	Singapore	51	1962:2012	21.96
Cuba	8	1999:2006	36.96	Slovenia	20	1992:2012	7.14
Cyprus	40	1972:2012	9.91	Spain	51	1962:2012	8.81
Czech Republic	20	1993:2012	7.33	St. Kitts and Nevis	23	1981:2011	0.16
Denmark	51	1962:2012	9.12	St. Lucia	35	1973:2008	0.68
Dominica	26	1977:2012	2.45	St. Vincent and the Grenadines	22	1976:2012	0.22
Equatorial Guinea	4	1980:1983	34.76	Suriname	24	1965:2011	10.38
Estonia	18	1995:2012	18.94	Sweden	51	1962:2012	16.07
Faeroe Islands	5	1998:2009	4.58	Switzerland	51	1962:2012	4.42
Fiji	41	1967:2012	2.74	Tonga	9	1975:2012	3.55
Finland	50	1963:2012	19.33	Turkmenistan	4	1997:2000	88.04
France	51	1962:2012	8.06	Turks and Caicos Islands	8	1999:2012	1.91
French Polynesia	30	1962:2012	1.70	United Arab Emirates	13	1978:2008	78.51
Germany	51	1962:2012	6.57	United Kingdom	51	1962:2012	11.04
Greenland	33	1978:2012	5.60	United States	51	1962:2012	11.41
Grenada	27	1977:2008	0.05	Virgin Islands (U.S.)	6	1970:1978	75.01
Hong Kong SAR, China	51	1962:2012	2.05				
Iceland	25	1988:2012	20.10				
Iran, Islamic Rep.	27	1963:2011	92.25				
Ireland	50	1963:2012	4.11				
Israel	51	1962:2012	4.86				
Italy	51	1962:2012	7.27				

g. Trade Among LAC4 Countries

Table A17. Trade among LAC4 Countries

Exporting Country	Importing country							
	Brazil	Chile	Colombia	Peru	LAC4	US	Others	Non-LAC4
Brazil		0.002	0.001	0.001	0.005	0.018	0.083	0.101
Chile	0.015		0.003	0.006	0.024	0.04	0.234	0.274
Colombia	0.002	0.003		0.004	0.008	0.051	0.072	0.124
Peru	0.006	0.009	0.004		0.019	0.042	0.141	0.183

Note: This table shows the trade among LAC4. Columns report the destination for every country in LAC4's exports as a share of total exports.

2. Model: Full Set of Equilibrium Conditions, Equilibrium Definition & Measurement Equations

a. Equilibrium Conditions in Levels & log-linearized

$$p_t^c C_t + p_t^x X_t + R_{t-1} D_{t-1} = w_t L_t + r_t^k K_{t-1} + D_t + p_t^{Co} \bar{Co} \quad (1)$$

$$\lambda_t p_t^c = \left(C_t - \psi^c \frac{L_t^{1+\gamma}}{1+\gamma} \right)^{-\sigma} \quad (2)$$

$$p_t^c \psi^c L_t^\gamma = w_t \quad (3)$$

$$\lambda_t = \beta R_t E_t \lambda_{t+1} \quad (4)$$

$$Q_t^x = \beta E_t \left\{ \frac{\lambda_{t+1}}{\lambda_t} \left(r_{t+1}^k + Q_{t+1}^x (1 - \delta) \right) \right\} \quad (5)$$

$$p_t^x = Q_t^x \left[1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} - 2 \right) - \frac{\sqrt{a}}{2} \left(\frac{X_t}{X_{t-1}} \right) \left(e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} - e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} \right) \right] - \beta E_t \left\{ \left(\frac{\lambda_{t+1}}{\lambda_t} \right) Q_{t+1}^x \frac{\sqrt{a}}{2} \left(\frac{X_{t+1}}{X_t} \right)^2 \left(e^{\left(\sqrt{a} \left(\frac{X_{t+1}}{X_t} - 1 \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{X_{t+1}}{X_t} - 1 \right) \right)} \right) \right\} \quad (6)$$

$$K_t = (1 - \delta) K_{t-1} + X_t \left[1 - s \left(e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} - 2 \right) \right] \quad (7)$$

$$C_t = \left[(1 - \alpha_c)^{\frac{1}{\eta_c}} \left(C_t^h \right)^{\frac{\eta_c - 1}{\eta_c}} + \alpha_c^{\frac{1}{\eta_c}} \left(C_t^f \right)^{\frac{\eta_c - 1}{\eta_c}} \right]^{\frac{\eta_c}{\eta_c - 1}} \quad (8)$$

$$C_t^h = (1 - \alpha_c) \left(\frac{p_t^h}{p_t^c} \right)^{-\eta_c} C_t \quad (9)$$

$$C_t^f = \alpha_c \left(\frac{1}{p_t^c} \right)^{-\eta_c} C_t \quad (10)$$

$$Y_t = z_t K_{t-1}^\alpha L_t^{1-\alpha} \quad (11)$$

$$w_t = p_t^h (1 - \alpha) \frac{Y_t}{L_t} \quad (12)$$

$$r_t^k = p_t^h \alpha \frac{Y_t}{K_{t-1}} \quad (13)$$

$$X_t = \left[(1 - \alpha_x)^{\frac{1}{\eta_x}} \left(X_t^h \right)^{\frac{\eta_x - 1}{\eta_x}} + \alpha_x^{\frac{1}{\eta_x}} \left(X_t^f \right)^{\frac{\eta_x - 1}{\eta_x}} \right]^{\frac{\eta_x}{\eta_x - 1}} \quad (14)$$

$$X_t^h = (1 - \alpha_x) \left(\frac{p_t^h}{p_t^x} \right)^{-\eta_x} X_t \quad (15)$$

$$X_t^f = \alpha_x \left(\frac{1}{p_t^x} \right)^{-\eta_x} X_t \quad (16)$$

$$Y_t = C_t^h + X_t^h + C_t^{h*} \quad (17)$$

$$GDP_t = p_t^h Y_t + p_t^{Co} \overline{Co} \quad (18)$$

$$RER_t = \frac{1}{p_t^c} \quad (19)$$

$$TB_t = \frac{p_t^h C_t^{h*} + p_t^{Co} \overline{Co} - C_t^f - X_t^f}{p_t^c} \quad (20)$$

$$C_{j,t}^{h*} = \left(p_{j,t}^h \right)^{-\epsilon_{j,e}} Y_t^* \quad (21)$$

$$R_{j,t} = R_{j,t}^* \quad (22)$$

$$R_{jt} = R_t^* S_{jt} \quad (23)$$

$$S_{jt} = z_{jt}^R \{ \exp [\Omega_{j,u}(D_{j,t} - \overline{D}_j)] - 1 \} \quad (24)$$

$$f_t^{Co} = \phi_{Co} f_{t-1}^{Co} + \sigma^f f_t^{Co} \varepsilon_t^{fCo} \quad (25)$$

$$p_{j,t}^{Co} = \omega_j^{Co} f_t^{Co} + \varepsilon_{j,t}^{Co} \quad (26)$$

$$\varepsilon_{j,t}^{Co} = \rho^{Co} \varepsilon_{j,t-1}^{Co} + \sigma_j^{Co} \nu_{j,t}^{Co} \quad (27)$$

$$\ln R_t^* = \rho_{R^*} \ln R_{t-1}^* + (1 - \rho_{R^*}) \ln \overline{R}^* + \sigma^{R^*} \varepsilon_t^{R^*} \quad (28)$$

$$z_{j,t}^R = \left(1 - \rho_j^R \right) \overline{z}_j^R + \rho_j^R z_{j,t-1}^R + \sigma_j^R \nu_{j,t}^R \quad (29)$$

$$\ln Y_t^* = (1 - \rho_{Y^*}) \ln \overline{Y}^* + \rho_{Y^*} \ln Y_{t-1}^* + \sigma^{Y^*} \varepsilon_t^{Y^*} \quad (30)$$

$$\ln z_{j,t} = (1 - \rho_j^z) \ln \overline{z}_j + \rho_j^z \ln z_{j,t-1} + \sigma_j^z \varepsilon_{j,t}^z \quad (31)$$

Full set of log-linearized equilibrium conditions

$$p^c C \left(\widehat{p}_t^c + \widehat{C}_t \right) + p^x X \left(\widehat{p}_t^x + \widehat{X}_t \right) + RD \left(\widehat{R}_{t-1} + \widehat{D}_{t-1} \right) = \\ wL \left(\widehat{w}_t + \widehat{L}_t \right) + r^k K \left(\widehat{r}_t^k + \widehat{K}_{t-1} \right) + D \widehat{D}_t + p^{Co} \overline{Co} \widehat{p}_t^{Co} \quad (32)$$

$$\widehat{\lambda}_t + \widehat{p}_t^c = -\sigma \left(\lambda^h p^c \right)^{\frac{1}{\sigma}} \left(C \widehat{C}_t - \psi^c H^{1+\gamma} \widehat{H}_t \right) \quad (33)$$

$$\gamma \widehat{L}_t = \widehat{w}_t - \widehat{p}_t^c \quad (34)$$

$$\widehat{\lambda}_t = \widehat{R}_t + E_t \left\{ \widehat{\lambda}_{t+1} \right\} \quad (35)$$

$$\widehat{Q}_t^x = E_t \left\{ \lambda_{t+1} - \lambda_t \right\} + \beta E_t \left\{ \frac{r^k}{Q^x} \widehat{r}_{t+1}^k + (1 - \delta) \widehat{Q}_{t+1}^x \right\} \quad (36)$$

$$\widehat{X}_t = \left(\frac{1}{(1 + \beta)a} \right) \left(\widehat{Q}_t^x - \widehat{p}_t^x \right) + \left(\frac{1}{1 + \beta} \right) \widehat{X}_{t-1} + \left(\frac{\beta}{1 + \beta} \right) \widehat{X}_{t+1} \quad (37)$$

$$\widehat{K}_t = (1 - \delta) \widehat{K}_{t-1} + \delta \widehat{X}_t \quad (38)$$

$$\widehat{C}_t = (1 - \alpha_c)^{\frac{1}{\eta_c}} \left(\frac{C^h}{C} \right)^{\frac{\eta_c - 1}{\eta_c}} \widehat{C}_t^h + \alpha_c^{\frac{1}{\eta_c}} \left(\frac{C^f}{C} \right)^{\frac{\eta_c - 1}{\eta_c}} \widehat{C}_t^f \quad (39)$$

$$\widehat{C}_t^h = -\eta_c \left(\widehat{p}_t^h - \widehat{p}_t^c \right) + \widehat{C}_t \quad (40)$$

$$\widehat{C}_t^f = \eta_c \widehat{p}_t^c + \widehat{C}_t \quad (41)$$

$$\widehat{Y}_t = \widehat{z}_t + \alpha \widehat{K}_{t-1} + (1 - \alpha) \widehat{L}_t \quad (42)$$

$$\widehat{w}_t = \widehat{p}_t^h + \widehat{Y}_t - \widehat{L}_t \quad (43)$$

$$\widehat{r}_t^k = \widehat{p}_t^h + \widehat{Y}_t - \widehat{K}_{t-1} \quad (44)$$

$$\widehat{X}_t = (1 - \alpha_x)^{\frac{1}{\eta_x}} \left(\frac{X^h}{X} \right)^{\frac{\eta_x - 1}{\eta_x}} \widehat{X}_t^h + \alpha_x^{\frac{1}{\eta_c}} \left(\frac{X^f}{X} \right)^{\frac{\eta_x - 1}{\eta_x}} \widehat{X}_t^f \quad (45)$$

$$\widehat{X}_t^h = -\eta_x \left(\widehat{p}_t^h - \widehat{p}_t^x \right) + \widehat{X}_t \quad (46)$$

$$\widehat{X}_t^f = \eta_x p_t^x + \widehat{X}_t \quad (47)$$

$$\widehat{Y}_t = \left(\frac{C^h}{Y} \right) \widehat{C}_t^h + \left(\frac{X^h}{Y} \right) \widehat{X}_t^h + \left(\frac{C^{h*}}{Y} \right) \widehat{C}_t^{h*} \quad (48)$$

$$\widehat{GDP}_t = \frac{P^h Y}{GDP} \left(\widehat{p}_t^h + \widehat{Y}_t \right) + \frac{p^{Co} \overline{Co}}{GDP} \widehat{p}_t^{Co} \quad (49)$$

$$\widehat{RE} \widehat{R}_t = \frac{1}{\widehat{p}_t^c} \quad (50)$$

$$\widehat{TB}_t = \frac{1}{p^c TB} \left(p^h C^{h*} \left(\widehat{p}_t^h + \widehat{C}_t^{h*} \right) + p^{Co} \overline{Co} \widehat{p}_t^{Co} - C^f \widehat{C}_t^f - X^f \widehat{X}_t^f - \widehat{p}_t^c \right) \quad (51)$$

$$\widehat{C}_{j,t}^{h*} = -\epsilon_{j,e} \widehat{p}_{j,t}^h + \widehat{Y}_t^* \quad (52)$$

$$\widehat{R}_{j,t} = \widehat{R}_{j,t}^* \quad (53)$$

$$\widehat{R}_{jt} = \widehat{R}_{jt}^* + \widehat{S}_{jt} \quad (54)$$

$$\widehat{S}_{jt} = \widehat{z}_{jt}^R + \Omega_{j,u} D_j \widehat{D}_{j,t} \quad (55)$$

$$\widehat{f}_t^{Co} = \phi_{Co} \widehat{f}_{t-1}^{Co} + \sigma^f \varepsilon_t^{Co} \quad (56)$$

$$\widehat{p}_{j,t}^{Co} = \omega_j^{Co} \widehat{f}_t^{Co} + \varepsilon_{j,t}^{Co} \quad (57)$$

$$\widehat{\varepsilon}_{j,t}^{Co} = \rho^{Co} \widehat{\varepsilon}_{j,t-1}^{Co} + \sigma_j^{Co} \nu_{j,t}^{Co} \quad (58)$$

$$\widehat{R}_t^* = \rho_{R^*} \widehat{R}_{t-1}^* + \sigma^{R^*} \varepsilon_t^{R^*} \quad (59)$$

$$\widehat{z}_{j,t}^R = \rho_j^R \widehat{z}_{j,t-1}^R + \sigma_j^R \nu_{j,t}^R \quad (60)$$

$$\widehat{Y}_t^* = \rho_{Y^*} \widehat{Y}_{t-1}^* + \sigma^{Y^*} \varepsilon_t^{Y^*} \quad (61)$$

$$\widehat{z}_{j,t} = \rho_j^z \widehat{z}_{j,t-1} + \sigma_j^z \varepsilon_{j,t}^z \quad (62)$$

b. Equilibrium Definition

Given initial conditions $K_{j,-1}$ and $D_{j,-1}$, and exogenous state-contingent sequences of idiosyncratic and common shocks $\{\nu_{j,t}^{Co}, \nu_{j,t}^R, \varepsilon_{j,t}^z, \varepsilon_t^{fCo}, \varepsilon_t^{R*}, \varepsilon_t^{Y*}\}$ in economy j , an equilibrium is a set of state-contingent allocations³

$$\{C_{j,t}, L_{j,t}, D_{j,t}, X_{j,t}, K_{j,t}, C_{j,t}^h, C_{j,t}^f, X_{j,t}^h, X_{j,t}^f, Y_{j,t}, C_{j,t}^{h*}\}_{t=0}^{\infty}$$

and prices

$$\{p_{j,t}^c, p_{j,t}^x, R_{j,t}^*, w_{j,t}, r_{j,t}^k\}_{t=0}^{\infty}$$

such that, for all j 's EMEs in the sample, given the laws of motion of shocks:

1. The allocations solve the consumer's problem given prices and the laws of motion for the capital stocks.
2. The allocations solve the firm's problem given prices.
3. Markets clear for capital, labor, investment goods, and home goods.

c. Measurement Equations

$$hp^c \left[\ln \left(\text{nominal } GDP_t^D / CPI_t^D \right) \right] = \ln (GDP_t) - \ln (\overline{GDP}) \quad (63)$$

$$hp^c \left[\ln \left(\text{real } C_t^D \right) \right] = \ln (C_t) - \ln (\overline{C}) \quad (64)$$

$$hp^c \left[\ln \left(\text{real } X_t^D \right) \right] = \ln (X_t) - \ln (\overline{X}) \quad (65)$$

$$hp^c \left[\ln \left(\text{real } ER_t^D \right) \right] = \ln (ER_t) - \ln (\overline{ER}) \quad (66)$$

$$hp^c \left[\ln \left(\text{real } Y_t^{*D} \right) \right] = \ln (Y_t^*) - \ln (\overline{Y^*}) \quad (67)$$

$$hp^c \left[\ln \left(\text{real } R_t^{*D} \right) \right] = \ln (R_t^*) - \ln (R^*) \quad (68)$$

$$hp^c \left[\ln \left(p_t^{Co,D} \right) \right] = \ln (p_t^{Co}) - \ln (\overline{p^{Co}}) \quad (69)$$

$$hp^c \left[\left(\text{Trade Balance}_t^D / \text{Output}_t^D \right) \right] = \left(\frac{TB_t}{GDP_t} \right) - \left(\frac{\overline{TB}}{\overline{GDP}} \right) \quad (70)$$

$$hp^c \left[\ln \left(1 + \frac{EMBI_t^D}{10000} \right)^{1/4} \right] = \ln (S_t) - \ln (\overline{S}) \quad (71)$$

3. Benchmark Estimation: Further results not shown in draft

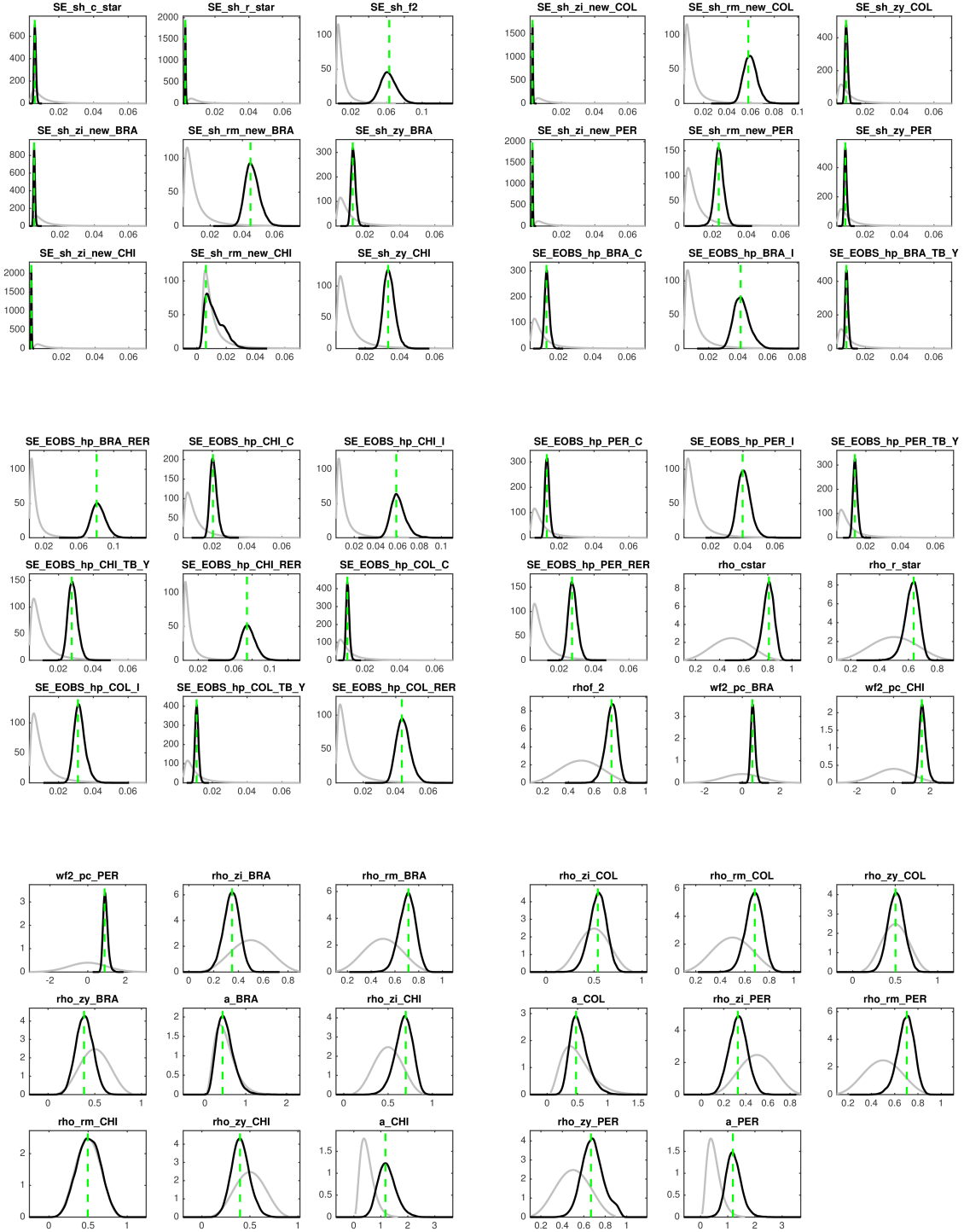
The identification analysis performed in Dynare 4.4.3⁴ shows that the necessary condition discussed in Iskrev (2010)⁵ is fulfilled. That is, Jacobian of the matrix that relates the parameters of the model with the model's solution has full rank, this implies that the vector of estimated parameters is locally identifiable.

³The set of equilibrium allocations and prices that characterize the solution of the model is found by applying a first-order Taylor approximation to the set of equilibrium conditions. The log-linearized system is then solved using perturbation methods in DYNARE.

⁴Stephane Adjemian, Houtan Bastani, Michel Juillard, Frederic Karame, Ferhat Mihoubi, George Perendia, Johannes Pfeifer, Marco Ratto and Sebastien Villemot (2011), "Dynare: Reference Manual, Version 4," Dynare Working Papers, 1, CEPREMAP

⁵Iskrev, N. (2010). Local identification in DSGE models. *Journal of Monetary Economics* 57, 189–202.

Figure A12. Benchmark Estimation: Priors and Posteriors



Note: This figure shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2.

Table A18. Standard deviation of measurement errors

s.d. of measurement errors	prior mean	prior	pstdev	mode	s.d.	post. mean	90% HPD interval	
Brazil								
c_t	0.013	invg	Inf	0.013	0.001	0.0132	0.0109	0.0153
x_t	0.013	invg	Inf	0.042	0.006	0.0423	0.0337	0.0512
$\frac{TB}{Y}$	0.013	invg	Inf	0.009	0.001	0.0091	0.0077	0.0105
RER_t	0.013	invg	Inf	0.080	0.008	0.0826	0.0698	0.0956
Chile								
c_t	0.013	invg	Inf	0.020	0.002	0.0207	0.0174	0.0239
x_t	0.013	invg	Inf	0.059	0.006	0.0604	0.0496	0.0713
$\frac{TB}{Y}$	0.013	invg	Inf	0.028	0.003	0.0282	0.0238	0.0326
RER_t	0.013	invg	Inf	0.074	0.007	0.0760	0.0630	0.0883
Colombia								
c_t	0.013	invg	Inf	0.010	0.001	0.0100	0.0085	0.0115
x_t	0.013	invg	Inf	0.031	0.003	0.0322	0.0271	0.0375
$\frac{TB}{Y}$	0.013	invg	Inf	0.011	0.001	0.0111	0.0095	0.0127
RER_t	0.013	invg	Inf	0.044	0.004	0.0446	0.0377	0.0514
Peru								
c_t	0.013	invg	Inf	0.013	0.001	0.0131	0.0111	0.0151
x_t	0.013	invg	Inf	0.040	0.004	0.0408	0.0342	0.0473
$\frac{TB}{Y}$	0.013	invg	Inf	0.014	0.001	0.0139	0.0119	0.0160
RER_t	0.013	invg	Inf	0.027	0.003	0.0278	0.0238	0.0319

Note: This table shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2.

Table A19. Second Moments: Data and Model

	S.D. (%)	Relative S.D. to GDP/CPI	Autocorrelation		Cross correlation with GDP/CPI ($t+j$)									
			1	2	$j = -4$	$j = -3$	$j = -2$	$j = -1$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$	
			Model											
P^{Co}	11.08	3.43	0.72	0.52	0.13	0.19	0.26	0.36	0.49	0.39	0.32	0.27	0.23	
GDP/CPI	3.80	1.00	0.79	0.66	0.53	0.59	0.66	0.79	1.00	0.79	0.66	0.59	0.53	
$Cons$	3.32	0.89	0.83	0.75	0.61	0.63	0.67	0.75	0.92	0.75	0.65	0.59	0.55	
Inv	7.40	2.00	0.91	0.80	0.55	0.61	0.69	0.77	0.80	0.64	0.54	0.49	0.46	
TB/GDP	1.57	0.40	0.76	0.57	-0.38	-0.36	-0.33	-0.26	-0.09	-0.13	-0.17	-0.19	-0.21	
RER	6.27	1.64	0.75	0.63	-0.23	-0.24	-0.24	-0.19	-0.01	-0.13	-0.18	-0.21	-0.22	
Data														
P^{Co}	12.14	4.24	0.75***	0.39***	0.11*	0.29***	0.49***	0.67***	0.73***	0.64***	0.45***	0.26***	0.10	
GDP/CPI	2.86	1.00	0.80***	0.56***	0.15**	0.36***	0.56***	0.80***	1.00***	0.80***	0.56***	0.36***	0.15**	
$Cons$	1.34	0.47	0.82***	0.60***	0.42***	0.56***	0.68***	0.72***	0.69***	0.53***	0.28***	0.06	-0.11**	
Inv	5.14	1.80	0.72***	0.42***	0.35***	0.45***	0.52***	0.54***	0.49***	0.30***	0.07	-0.05	-0.14**	
TB/GDP	1.99	0.70	0.68***	0.41***	-0.13*	-0.07	0.05	0.25***	0.46***	0.52***	0.52***	0.46***	0.31***	
RER	5.44	1.90	0.63***	0.19***	-0.09	-0.08	-0.11*	-0.22***	-0.27***	-0.30***	0.25***	-0.17**	-0.10	

Note: This table reports other moments of the data and the model for the commodity price index, nominal GDP over CPI, consumption, investment, trade balance over GDP and real exchange rate. The panels report the absolute standard deviation, the standard deviation relative to GDP/CPI, the autocorrelation coefficient of orders one and two and the cross correlation for every variable in period t with GDP/CPI in period $t+j$ with $j = -4, \dots, 4$. Statistical significance at 1, 5, and 10 percent is reported with (***), (**), and (*), respectively.

Table A20. Counterfactual Historical Decomposition

shocks	Brazil	Chile	Colombia	Peru
S.D. of observed real income cycle	0.0218	0.0418	0.0184	0.0273
Counterfactual S.D. excluding each shock separately (in percent of the observed S.D) (%)				
ν^R (Idiosyncratic spread)	92.7	98.8	97.7	99.9
ν^{Co} (Idiosyncratic commodity price)	103.1	99.7	98.2	94.0
ε^z (Domestic productivity)	53.0	114.9	55.1	43.9
ε^{R^*} (World interest rate)	105.6	101.2	100.4	99.8
ε^{Y^*} (World demand)	99.4	99.3	98.1	100.6
ε^{fCo} (Common factor in commodity prices)	89.1	72.2	80.3	75.4
Counterfactual S.D. excluding two commodities price shocks (%)				
$\{\nu^{Co}, \varepsilon^{fCo}\}$	91.1	72.1	75.4	69.6
Counterfactual S.D. excluding all shocks except commodities price shocks (%)				
$\{\nu^R, \varepsilon^z, \varepsilon^{R^*}, \varepsilon^{Y^*}\}$	41.5	114.7	47.9	44.1

Note: The table presents in rows two and below the counterfactual standard deviation (S.D) of real income implied by the model when one or several of the structural shocks are turned off. The numbers reported are in percentage of the observed S.D. presented in the first row computed over the entire sample period of estimation: 2000.Q1 to 2014.Q3.

4. Extensions & Robustness in Draft: Further results

Table A21. Comparison with an SVAR

	Brazil	Chile	Colombia	Peru	Median	Mean
A. Unconditional Variance Decomposition of Output: Share of Pcom shocks						
SVAR	15	53	47	80	50	49
DSGE	28	77	44	40	42	47
B. SD(Output-counterfactual)/SD(Output-data): Only Pcom shocks						
SVAR	41	74	63	91	69	67
DSGE	42	115	48	44	46	62
C. SD(Output-counterfactual)/SD(Output-data): All shocks except Pcom						
SVAR	88	90	71	48	80	74
DSGE	91	72	75	70	74	77
D. corr(Output-counterfactual,Output-data): Only Pcom shocks						
SVAR	0.45	0.55	0.71	0.88	0.63	0.65
DSGE	0.41	0.78	0.67	0.82	0.73	0.67
E. corr(Output-counterfactual-SVAR,Output-counterfactual-DSGE): Only Pcom shocks						
	0.88	0.27	0.96	0.91	0.89	0.75

Note: PCom stands for country-specific commodity price. SD(X) stands for the standard deviation of variable "X". This table reports some statistics that compare the estimated SVAR and the DSGE model. See the footnote in Figure 9 for further information on the SVAR estimation procedure. In the table, "counterfactual" stands for the simulated series from the model either when all shocks but the commodity price one are shut down or when only the commodity price shock is shut down.

Table A22. Countercyclical Spreads: Priors/Posteriors

Parameters	Prior			Posterior				
	Mean	Type	S.D.	Mode	S.D.	Mean	90% HPD interval	
Global								
$\rho_{Y^{\star}}$	0.500	beta	0.150	0.8022	0.0467	0.7983	0.7217	0.8768
$\rho_{R^{\star}}$	0.500	beta	0.150	0.6185	0.0495	0.6066	0.5266	0.6867
ϕ_{Co}	0.500	beta	0.150	0.7098	0.0473	0.6933	0.6219	0.7611
$\sigma^{Y^{\star}}$	0.013	invg	Inf	0.0061	0.0005	0.0062	0.0053	0.0072
$\sigma^{R^{\star}}$	0.013	invg	Inf	0.0024	0.0002	0.0025	0.0021	0.0028
$\sigma^{f^{Co}}$	0.013	invg	Inf	0.0629	0.0097	0.0618	0.0465	0.0766
Brazil								
ρ^{zR}	0.50	beta	0.150	0.3178	0.0638	0.3088	0.2067	0.4170
ρ^{Co}	0.50	beta	0.150	0.6767	0.0686	0.6551	0.5534	0.7600
ρ^z	0.50	beta	0.15	0.3731	0.1018	0.3816	0.2322	0.5369
a	0.50	gamma	0.250	0.4766	0.2031	0.5425	0.1944	0.8588
η^z	0.05	gamm	0.03	0.0337	0.0193	0.0449	0.0106	0.0775
η^{Co}	0.05	gamm	0.03	0.0226	0.0085	0.0241	0.0102	0.0370
ω^{Co}	0.00	norm	1.0	0.5300	0.1062	0.5577	0.3629	0.7453
ψ^H	0.50	beta	0.15	0.1865	0.0963	0.2168	0.0714	0.3512
σ^{zR}	0.01	invg	Inf	0.0049	0.0005	0.0051	0.0043	0.0058
σ^{Co}	0.01	invg	Inf	0.0451	0.0044	0.0464	0.0394	0.0531
σ^z	0.01	invg	Inf	0.0131	0.0011	0.0136	0.0114	0.0158
c_t	0.013	invg	Inf	0.0132	0.0013	0.0136	0.0113	0.0159
x_t	0.013	invg	Inf	0.0409	0.0059	0.0414	0.0327	0.0498
$\frac{TB}{Y}$	0.013	invg	Inf	0.0086	0.0008	0.0088	0.0075	0.0101
RER_t	0.013	invg	Inf	0.0792	0.0075	0.0811	0.0686	0.0936
Chile								
ρ^{zR}	0.50	beta	0.150	0.6675	0.1144	0.6489	0.4742	0.8080
ρ^{Co}	0.50	beta	0.150	0.4969	0.1427	0.5098	0.2651	0.7600
ρ^z	0.50	beta	0.150	0.4014	0.0730	0.3955	0.2676	0.5222
a	0.50	gamma	0.250	1.1537	0.3432	1.1890	0.6556	1.6798
η^z	0.05	gamm	0.03	0.0571	0.0158	0.0616	0.0297	0.0921
η^{Co}	0.05	gamm	0.03	0.0043	0.0020	0.0052	0.0019	0.0083
ω^{Co}	0.00	norm	1.0	1.5424	0.1821	1.6240	1.2985	1.9393
ψ^H	0.50	beta	0.15	0.4995	0.1497	0.4988	0.2619	0.7548
σ^{zR}	0.01	invg	Inf	0.0020	0.0002	0.0020	0.0017	0.0024
σ^{Co}	0.01	invg	Inf	0.0061	0.0074	0.0124	0.0033	0.0231
σ^z	0.01	invg	Inf	0.0330	0.0030	0.0336	0.0287	0.0386
c_t	0.013	invg	Inf	0.0205	0.0021	0.0211	0.0179	0.0244
x_t	0.013	invg	Inf	0.0595	0.0061	0.0606	0.0499	0.0710
$\frac{TB}{Y}$	0.013	invg	Inf	0.0278	0.0028	0.0287	0.0243	0.0330
RER_t	0.013	invg	Inf	0.0728	0.0075	0.0739	0.0618	0.0860

Note: This table shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2. HPD stands for higher posterior density.

Table A22. Countercyclical Spreads: Priors/Posteriors (cont.)

Parameters	Prior			Posterior				
	Mean	Type	S.D.	Mode	S.D.	Mean	90% HPD interval	
Colombia								
ρ^{zR}	0.50	beta	0.150	0.4616	0.0996	0.4506	0.2949	0.6036
ρ^{Co}	0.50	beta	0.150	0.6305	0.0685	0.6149	0.5063	0.7285
ρ^z	0.50	beta	0.150	0.4689	0.0941	0.4649	0.3107	0.6076
a	0.50	gamma	0.250	0.5165	0.1584	0.5667	0.3061	0.8071
η^z	0.05	gamm	0.03	0.0279	0.0206	0.0367	0.0090	0.0637
η^{Co}	0.05	gamm	0.03	0.0077	0.0028	0.0088	0.0037	0.0139
ψ^H	0.50	beta	0.15	0.3853	0.1433	0.4090	0.1773	0.6398
σ^{zR}	0.01	invg	Inf	0.0026	0.0003	0.0027	0.0023	0.0031
σ^{Co}	0.01	invg	Inf	0.0586	0.0054	0.0607	0.0512	0.0706
σ^z	0.01	invg	Inf	0.0088	0.0009	0.0092	0.0077	0.0105
c_t	0.013	invg	Inf	0.0099	0.0010	0.0101	0.0086	0.0117
x_t	0.013	invg	Inf	0.0312	0.0032	0.0320	0.0270	0.0368
$\frac{TB}{Y}$	0.013	invg	Inf	0.0109	0.0012	0.0111	0.0093	0.0127
RER_t	0.013	invg	Inf	0.0430	0.0044	0.0441	0.0370	0.0509
Peru								
ρ^{zR}	0.50	beta	0.150	0.2365	0.0751	0.2439	0.1173	0.3578
ρ^{Co}	0.50	beta	0.150	0.7003	0.0689	0.6866	0.5705	0.8006
ρ^z	0.50	beta	0.150	0.6234	0.0983	0.6130	0.4692	0.7530
a	0.50	gamma	0.250	1.2736	0.2818	1.3296	0.8627	1.7984
η^z	0.05	gamm	0.03	0.0228	0.0155	0.0299	0.0067	0.0518
η^{Co}	0.05	gamm	0.03	0.0054	0.0026	0.0064	0.0021	0.0105
ω^{Co}	0.00	norm	1.0	0.8960	0.1147	0.9632	0.7518	1.1588
ψ^H	0.50	beta	0.15	0.4224	0.1683	0.4428	0.2081	0.6805
σ^{zR}	0.01	invg	Inf	0.0023	0.0002	0.0023	0.0020	0.0027
σ^{Co}	0.01	invg	Inf	0.0236	0.0028	0.0233	0.0187	0.0283
σ^z	0.01	invg	Inf	0.0082	0.0008	0.0084	0.0071	0.0096
c_t	0.013	invg	Inf	0.0130	0.0010	0.0132	0.0112	0.0151
x_t	0.013	invg	Inf	0.0401	0.0037	0.0412	0.0340	0.0480
$\frac{TB}{Y}$	0.013	invg	Inf	0.0137	0.0013	0.0140	0.0117	0.0161
RER_t	0.013	invg	Inf	0.0274	0.0031	0.0280	0.0239	0.0323

Note: This table shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2. HPD stands for higher posterior density.

Table A23. Correlation between external forces: Priors/Posteriors

Parameters	Prior			Posterior				
	Mean	Type	S.D.	Mode	S.D.	Mean	90% HPD interval	
Global								
ρ_{Y^*}	0.500	beta	0.150	0.811	0.045	0.805	0.732	0.878
ρ_{R^*}	0.500	beta	0.150	0.633	0.046	0.620	0.542	0.701
ϕ_{Co}	0.500	beta	0.150	0.708	0.050	0.706	0.624	0.785
σ^{Y^*}	0.013	inv	Inf	0.006	0.001	0.006	0.005	0.007
σ^{R^*}	0.013	inv	Inf	0.002	0.000	0.003	0.002	0.003
σ^{fCo}	0.013	inv	Inf	0.055	0.008	0.056	0.042	0.070
σ^{Co, Y^*}	0.013	inv	Inf	0.021	0.009	0.021	0.007	0.033
Brazil								
ρ^{zR}	0.50	beta	0.150	0.350	0.063	0.345	0.249	0.448
ρ^{Co}	0.50	beta	0.150	0.714	0.072	0.704	0.588	0.815
ρ^z	0.50	beta	0.15	0.386	0.092	0.393	0.246	0.542
a	0.50	gamma	0.250	0.430	0.194	0.496	0.175	0.812
ω^{Co}	0.00	norm	1.0	0.549	0.111	0.568	0.369	0.756
σ^{zR}	0.01	inv	Inf	0.005	0.000	0.005	0.004	0.006
σ^{Co}	0.01	inv	Inf	0.045	0.004	0.046	0.039	0.053
σ^z	0.01	inv	Inf	0.013	0.001	0.013	0.011	0.015
c_t	0.013	inv	Inf	0.013	0.001	0.013	0.011	0.015
x_t	0.013	inv	Inf	0.042	0.006	0.042	0.034	0.051
$\frac{TB}{Y}$	0.013	inv	Inf	0.009	0.001	0.009	0.008	0.011
RER_t	0.013	inv	Inf	0.081	0.008	0.083	0.069	0.096
Chile								
ρ^{zR}	0.50	beta	0.150	0.716	0.095	0.691	0.538	0.850
ρ^{Co}	0.50	beta	0.150	0.493	0.174	0.494	0.254	0.733
ρ^z	0.50	beta	0.150	0.398	0.093	0.404	0.245	0.553
a	0.50	gamma	0.250	1.153	0.312	1.230	0.685	1.739
ω^{Co}	0.00	norm	1.0	1.596	0.184	1.633	1.313	1.961
σ^{zR}	0.01	inv	Inf	0.002	0.000	0.002	0.002	0.003
σ^{Co}	0.01	inv	Inf	0.006	0.003	0.012	0.003	0.023
σ^z	0.01	inv	Inf	0.033	0.003	0.034	0.029	0.039
c_t	0.013	inv	Inf	0.021	0.002	0.021	0.018	0.024
x_t	0.013	inv	Inf	0.059	0.006	0.060	0.050	0.070
$\frac{TB}{Y}$	0.013	inv	Inf	0.028	0.003	0.029	0.024	0.033
RER_t	0.013	inv	Inf	0.074	0.007	0.075	0.063	0.087

Note: This table shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2. HPD stands for higher posterior density.

Table A23. Correlation between external forces: Priors/Posteriors (cont.)

Parameters	Prior			Posterior				
	Mean	Type	S.D.	Mode	S.D.	Mean	90% HPD interval	
Colombia								
ρ^{zR}	0.50	beta	0.150	0.541	0.093	0.532	0.383	0.677
ρ^{Co}	0.50	beta	0.150	0.680	0.069	0.672	0.559	0.789
ρ^z	0.50	beta	0.150	0.504	0.093	0.509	0.359	0.658
a	0.50	gamma	0.250	0.468	0.131	0.516	0.273	0.744
σ^{zR}	0.01	invg	Inf	0.003	0.000	0.003	0.002	0.003
σ^{Co}	0.01	invg	Inf	0.058	0.005	0.060	0.050	0.069
σ^z	0.01	invg	Inf	0.009	0.001	0.009	0.008	0.010
c_t	0.013	invg	Inf	0.010	0.001	0.010	0.009	0.012
x_t	0.013	invg	Inf	0.031	0.003	0.032	0.027	0.037
$\frac{TB}{Y}$	0.013	invg	Inf	0.011	0.001	0.011	0.010	0.013
RER_t	0.013	invg	Inf	0.044	0.004	0.045	0.038	0.052
Peru								
ρ^{zR}	0.50	beta	0.150	0.334	0.085	0.337	0.197	0.472
ρ^{Co}	0.50	beta	0.150	0.704	0.066	0.693	0.577	0.813
ρ^z	0.50	beta	0.150	0.668	0.092	0.693	0.519	0.877
a	0.50	gamma	0.250	1.154	0.253	1.166	0.691	1.632
ω^{Co}	0.00	norm	1.0	0.928	0.117	0.964	0.754	1.162
σ^{zR}	0.01	invg	Inf	0.002	0.000	0.002	0.002	0.003
σ^{Co}	0.01	invg	Inf	0.024	0.002	0.023	0.019	0.028
σ^z	0.01	invg	Inf	0.008	0.001	0.008	0.007	0.010
c_t	0.013	invg	Inf	0.013	0.001	0.013	0.011	0.015
x_t	0.013	invg	Inf	0.040	0.004	0.041	0.034	0.047
$\frac{TB}{Y}$	0.013	invg	Inf	0.014	0.001	0.014	0.012	0.016
RER_t	0.013	invg	Inf	0.027	0.003	0.028	0.024	0.032

Note: This table shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2.

Table A24. No Commodity Endowment: Priors/Posteriors

Parameters	Prior			Posterior				
	Mean	Type	S.D.	Mode	S.D.	Mean	90% HPD interval	
Global								
ρ_{Y^*}	0.500	beta	0.150	0.805	0.047	0.798	0.719	0.875
ρ_{R^*}	0.500	beta	0.150	0.599	0.053	0.586	0.498	0.679
ϕ_{Co}	0.500	beta	0.150	0.657	0.077	0.701	0.585	0.813
σ^{Y^*}	0.013	invga	Inf	0.006	0.001	0.006	0.005	0.007
σ^{R^*}	0.013	invga	Inf	0.002	0.000	0.003	0.002	0.003
$\sigma^{f^{Co}}$	0.013	invga	Inf	0.086	0.008	0.063	0.047	0.078
Brazil								
ρ^{zR}	0.50	beta	0.150	0.374	0.062	0.365	0.266	0.466
ρ^{Co}	0.50	beta	0.150	0.667	0.078	0.669	0.547	0.806
ρ^z	0.50	beta	0.15	0.443	0.091	0.447	0.300	0.601
a	0.50	gamma	0.250	0.473	0.206	0.527	0.198	0.846
ω^{Co}	0.00	norm	1.0	0.451	0.062	0.554	0.370	0.740
σ^{zR}	0.01	invga	Inf	0.005	0.000	0.005	0.004	0.006
σ^{Co}	0.01	invga	Inf	0.041	0.004	0.047	0.039	0.053
σ^z	0.01	invga	Inf	0.013	0.001	0.014	0.011	0.016
c_t	0.013	invga	Inf	0.013	0.001	0.013	0.011	0.015
x_t	0.013	invga	Inf	0.038	0.005	0.038	0.030	0.047
$\frac{TB}{Y}$	0.013	invga	Inf	0.009	0.001	0.009	0.007	0.010
RER_t	0.013	invga	Inf	0.080	0.008	0.083	0.070	0.096
Chile								
ρ^{zR}	0.50	beta	0.150	0.546	0.119	0.552	0.366	0.744
ρ^{Co}	0.50	beta	0.150	0.726	0.066	0.513	0.266	0.753
ρ^z	0.50	beta	0.150	0.978	0.007	0.970	0.950	0.988
a	0.50	gamma	0.250	0.992	0.366	1.056	0.430	1.658
ω^{Co}	0.00	norm	1.0	0.845	0.099	1.615	1.316	1.936
σ^{zR}	0.01	invga	Inf	0.002	0.000	0.002	0.002	0.002
σ^{Co}	0.01	invga	Inf	0.066	0.006	0.011	0.003	0.021
σ^z	0.01	invga	Inf	0.026	0.002	0.026	0.022	0.030
c_t	0.013	invga	Inf	0.022	0.003	0.021	0.017	0.025
x_t	0.013	invga	Inf	0.064	0.009	0.071	0.052	0.090
$\frac{TB}{Y}$	0.013	invga	Inf	0.029	0.003	0.030	0.025	0.035
RER_t	0.013	invga	Inf	0.109	0.010	0.110	0.092	0.126

Note: This table shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2. HPD stands for higher posterior density.

Table A24. No Commodity Endowment: Priors/Posteriors (cont.)

Parameters	Prior			Posterior				
	Mean	Type	S.D.	Mode	S.D.	Mean	90% HPD interval	
Colombia								
ρ^{zR}	0.50	beta	0.150	0.434	0.094	0.425	0.277	0.578
ρ^{Co}	0.50	beta	0.150	0.514	0.179	0.683	0.564	0.804
ρ^z	0.50	beta	0.150	0.571	0.083	0.579	0.440	0.724
a	0.50	gamma	0.250	0.786	0.192	0.841	0.500	1.169
σ^{zR}	0.01	invG	Inf	0.003	0.000	0.003	0.002	0.003
σ^{Co}	0.01	invG	Inf	0.007	0.004	0.060	0.051	0.069
σ^z	0.01	invG	Inf	0.010	0.001	0.010	0.009	0.012
c_t	0.013	invG	Inf	0.011	0.001	0.011	0P.009	0.013
x_t	0.013	invG	Inf	0.033	0.003	0.034	0.028	0.039
$\frac{TB}{Y}$	0.013	invG	Inf	0.012	0.001	0.012	0.010	0.014
RER_t	0.013	invG	Inf	0.053	0.005	0.054	0.046	0.063
Peru								
ρ^{zR}	0.50	beta	0.150	0.400	0.092	0.400	0.252	0.550
ρ^{Co}	0.50	beta	0.150	0.772	0.058	0.707	0.587	0.827
ρ^z	0.50	beta	0.150	0.775	0.082	0.767	0.649	0.890
a	0.50	gamma	0.250	1.181	0.276	1.216	0.726	1.689
ω^{Co}	0.00	norm	1.0	0.476	0.068	0.951	0.763	1.154
σ^{zR}	0.01	invG	Inf	0.002	0.000	0.002	0.002	0.003
σ^{Co}	0.01	invG	Inf	0.045	0.004	0.024	0.019	0.028
σ^z	0.01	invG	Inf	0.010	0.001	0.010	0.009	0.012
c_t	0.013	invG	Inf	0.019	0.002	0.019	0.016	0.022
x_t	0.013	invG	Inf	0.045	0.004	0.046	0.039	0.053
$\frac{TB}{Y}$	0.013	invG	Inf	0.021	0.002	0.021	0.018	0.024
RER_t	0.013	invG	Inf	0.038	0.004	0.039	0.033	0.045

Note: This table shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2.

Table A25. No Commodity Endowment: Second Moments

	S.D. (%)	Relative S.D. to GDP/CPI	Autocorrelation		Cross correlation with GDP/CPI ($t+j$)									
			1	2	$j = -4$	$j = -3$	$j = -2$	$j = -1$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$	
Model														
p^{Co}	10.86	3.36	0.69	0.48	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
GDP/CPI	4.66	1.00	0.82	0.71	0.58	0.63	0.71	0.82	1.00	0.82	0.71	0.63	0.58	
$Cons$	4.19	0.92	0.83	0.74	0.60	0.64	0.70	0.81	0.99	0.83	0.72	0.65	0.60	
Inv	7.19	1.66	0.90	0.77	0.48	0.58	0.69	0.80	0.85	0.70	0.61	0.54	0.50	
TB/GDP	0.68	0.20	0.90	0.78	-0.24	-0.33	-0.41	-0.46	-0.41	-0.36	-0.33	-0.31	-0.29	
RER	9.46	1.93	0.82	0.72	0.24	0.26	0.30	0.38	0.56	0.41	0.32	0.25	0.21	
Data														
p^{Co}	12.14	4.24	0.75***	0.39***	0.11*	0.29***	0.49***	0.67***	0.73***	0.64***	0.45***	0.26***	0.10	
GDP/CPI	2.86	1.00	0.80***	0.56***	0.15**	0.36***	0.56***	0.80***	1.00***	0.80***	0.56***	0.36***	0.15**	
$Cons$	1.34	0.47	0.82***	0.60***	0.42***	0.56***	0.68***	0.72***	0.69***	0.53***	0.28***	0.06	-0.11**	
Inv	5.14	1.80	0.72***	0.42***	0.35***	0.45***	0.52***	0.54***	0.49***	0.30***	0.07	-0.05	-0.14**	
TB/GDP	1.99	0.70	0.68***	0.41***	-0.13*	-0.07	0.05	0.25***	0.46***	0.52***	0.52***	0.46***	0.31***	
RER	5.44	1.90	0.63***	0.19***	-0.09	-0.08	-0.11*	-0.22***	-0.27***	-0.30***	0.25***	-0.17**	-0.10	

Note: This table reports other moments of the data and the model without commodities for the commodity price index, nominal GDP over CPI, consumption, investment, trade balance over GDP and real exchange rate. The panels report the absolute standard deviation, the standard deviation relative to GDP/CPI, the autocorrelation coefficient of orders one and two and the cross correlation for every variable in period t with GDP/CPI in period $t+j$ with $j = -4, \dots, 4$. Statistical significance at 1, 5, and 10 percent is reported with (***), (**), and (*), respectively.

5. Extension using Growth Rates: Further Details and Results

Among the extensions considered in the main text, we explore the robustness of our results to considering using growth rates as an alternative filter. This section describes the details of this extension not reported in the main text. We report the modified measurement equations as well as the prior/posterior results.

a. Measurement Equations

$$hp^c \left[\ln \left(\text{nominal } GDP_t^D / CPI_t^D \right) \right] - \gamma_j^{\text{output}} = \ln (GDP_t) - \ln (GDP_{t-1}) \quad (72)$$

$$hp^c \left[\ln \left(\text{real } C_t^D \right) \right] - \gamma_j^{\text{consumption}} = \ln (C_t) - \ln (C_{t-1}) \quad (73)$$

$$hp^c \left[\ln \left(\text{real } X_t^D \right) \right] - \gamma_j^{\text{investment}} = \ln (X_t) - \ln (X_{t-1}) \quad (74)$$

$$hp^c \left[\ln \left(\text{real } ER_t^D \right) \right] - \gamma_j^{\text{real exchange rate}} = \ln (RER_t) - \ln (RER_{t-1}) \quad (75)$$

$$hp^c \left[\ln \left(\text{real } Y_t^{*D} \right) \right] - \gamma_j^{\text{foreign output}} = \ln (Y_t^*) - \ln (Y_{t-1}^*) \quad (76)$$

$$hp^c \left[\ln \left(\text{real } R_t^{*D} \right) \right] - \gamma_j^{\text{foreign output}} = \ln (R_t^*) - \ln (R_{t-1}^*) \quad (77)$$

$$hp^c \left[\ln \left(p_t^{Co,D} \right) \right] - \gamma_j^{\text{commodity price}} = \ln (p_t^{co}) - \ln (p_{t-1}^{co}) \quad (78)$$

$$hp^c \left[\left(\text{Trade Balance}_t^D / \text{Output}_t^D \right) \right] - \gamma_j^{\text{trade balance over GDP}} = \left(\frac{TB_t}{GDP_t} \right) - \left(\frac{TB_{t-1}}{GDP_{t-1}} \right) \quad (79)$$

$$hp^c \left[\ln \left(1 + \frac{EMBI_t^D}{10000} \right)^{1/4} \right] - \gamma_j^{\text{EMBI}} = \ln (S_t) - \ln (S_{t-1}) \quad (80)$$

b. Priors/Posteriors: Parameters & Measurement Errors

Table A26. Growth Rates: Parameters and Measurement Errors

Parameters	Prior			Posterior				
	Mean	Type	S.D.	Mode	S.D.	Mean	90% HPD interval	
Global								
ρ_{Y^*}	0.500	beta	0.150	0.906	0.028	0.902	0.855	0.947
ρ_{R^*}	0.500	beta	0.150	0.581	0.057	0.577	0.484	0.668
ϕ_{Co}	0.500	beta	0.150	0.937	0.012	0.932	0.909	0.955
σ^{Y^*}	0.013	invg	Inf	0.007	0.001	0.007	0.006	0.008
σ^{R^*}	0.013	invg	Inf	0.003	0.000	0.003	0.002	0.003
$\sigma^{f^{Co}}$	0.013	invg	Inf	0.065	0.008	0.066	0.052	0.081
Brazil								
ρ^{zR}	0.50	beta	0.150	0.536	0.048	0.529	0.448	0.614
ρ^{Co}	0.50	beta	0.150	0.843	0.038	0.842	0.776	0.905
ρ^z	0.50	beta	0.15	0.595	0.091	0.596	0.448	0.748
a	0.50	gamma	0.250	0.519	0.293	0.630	0.179	1.079
ω^{Co}	0.00	norm	1.0	0.668	0.104	0.674	0.482	0.868
σ^{zR}	0.01	invg	Inf	0.005	0.000	0.005	0.004	0.006
σ^{Co}	0.01	invg	Inf	0.047	0.004	0.049	0.041	0.057
σ^z	0.01	invg	Inf	0.015	0.001	0.015	0.013	0.018
c_t	0.013	invg	Inf	0.017	0.002	0.017	0.014	0.020
x_t	0.013	invg	Inf	0.048	0.008	0.048	0.036	0.059
$\frac{TB}{Y}$	0.013	invg	Inf	0.008	0.001	0.008	0.007	0.009
RER_t	0.013	invg	Inf	0.072	0.007	0.074	0.062	0.087
Chile								
ρ^{zR}	0.50	beta	0.150	0.916	0.041	0.903	0.838	0.973
ρ^{Co}	0.50	beta	0.150	0.833	0.051	0.746	0.523	0.913
ρ^z	0.50	beta	0.150	0.943	0.013	0.939	0.915	0.963
a	0.50	gamma	0.250	1.003	0.335	1.128	0.568	1.675
ω^{Co}	0.00	norm	1.0	1.530	0.155	1.554	1.282	1.817
σ^{zR}	0.01	invg	Inf	0.002	0.000	0.002	0.002	0.003
σ^{Co}	0.01	invg	Inf	0.034	0.005	0.031	0.011	0.047
σ^z	0.01	invg	Inf	0.048	0.004	0.049	0.041	0.056
c_t	0.013	invg	Inf	0.013	0.001	0.014	0.011	0.016
x_t	0.013	invg	Inf	0.043	0.004	0.043	0.036	0.050
$\frac{TB}{Y}$	0.013	invg	Inf	0.021	0.002	0.022	0.019	0.025
RER_t	0.013	invg	Inf	0.112	0.011	0.112	0.093	0.130

Note: This table shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2. HPD stands for higher posterior density.

Table A26. Growth Rates: Parameters and Measurement Errors (cont.)

Parameters	Prior			Posterior				
	Mean	Type	S.D.	Mode	S.D.	Mean	90% HPD interval	
Colombia								
ρ^{zR}	0.50	beta	0.150	0.711	0.064	0.698	0.595	0.808
ρ^{Co}	0.50	beta	0.150	0.706	0.072	0.712	0.604	0.825
ρ^z	0.50	beta	0.150	0.718	0.098	0.711	0.572	0.866
a	0.50	gamma	0.250	0.375	0.135	0.440	0.192	0.675
σ^{zR}	0.01	invg	Inf	0.003	0.000	0.003	0.002	0.003
σ^{Co}	0.01	invg	Inf	0.057	0.006	0.060	0.050	0.069
σ^z	0.01	invg	Inf	0.009	0.001	0.010	0.008	0.011
c_t	0.013	invg	Inf	0.010	0.001	0.010	0.008	0.011
x_t	0.013	invg	Inf	0.037	0.003	0.037	0.032	0.043
$\frac{TB}{Y}$	0.013	invg	Inf	0.012	0.001	0.013	0.011	0.014
RER_t	0.013	invg	Inf	0.032	0.003	0.032	0.028	0.037
Peru								
ρ^{zR}	0.50	beta	0.150	0.562	0.078	0.559	0.432	0.686
ρ^{Co}	0.50	beta	0.150	0.737	0.103	0.748	0.558	0.925
ρ^z	0.50	beta	0.150	0.850	0.059	0.834	0.748	0.930
a	0.50	gamma	0.250	0.950	0.264	1.026	0.563	1.450
ω^{Co}	0.00	norm	1.0	0.990	0.105	0.995	0.797	1.195
σ^{zR}	0.01	invg	Inf	0.002	0.000	0.002	0.002	0.003
σ^{Co}	0.01	invg	Inf	0.021	0.004	0.022	0.012	0.031
σ^z	0.01	invg	Inf	0.009	0.001	0.009	0.008	0.011
c_t	0.013	invg	Inf	0.015	0.001	0.015	0.013	0.017
x_t	0.013	invg	Inf	0.040	0.004	0.040	0.034	0.046
$\frac{TB}{Y}$	0.013	invg	Inf	0.014	0.001	0.014	0.012	0.016
RER_t	0.013	invg	Inf	0.031	0.003	0.031	0.026	0.036

Note: This table shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2.

6. Extension using Other Filters

This section presents the variance decomposition results of the model estimated with data filtered with two alternative detrending methods: (i) log-linear detrending; and (ii) log-quadratic detrending.

Table A27. Forecast Error Variance Decomposition of Output: Other Filters

shocks	Brazil	Chile	Colombia	Peru	Mean	Median	shocks	Brazil	Chile	Colombia	Peru	Mean	Median
Commodity price shock							Commodity price shock						
ν^{Co}	7.55	3.15	3.96	1.93	4.15	3.56	ν^{Co}	30.82	0.32	12.24	4.13	11.88	8.19
ε^{fCo}	23.86	64.25	73.48	71.47	58.27	67.86	ε^{fCo}	16.81	53.62	22.81	34.76	32.00	28.79
	31.41	67.40	77.44	73.40	62.41	71.42		47.63	53.94	35.05	38.89	43.88	36.97
All other shocks							All other shocks						
ν^R	3.87	0.21	1.45	0.08	1.40	0.83	ν^R	19.42	0.38	3.42	0.2	5.855	1.9
ε^z	61.15	32.37	16.69	26.06	34.07	29.22	ε^z	24.17	45.65	57.23	60.38	46.8575	51.44
ε^{R*}	3.29	0.01	1.38	0.22	1.23	0.80	ε^{R*}	8.24	0.01	1.75	0.27	2.5675	1.01
ε^{Y*}	0.28	0.02	3.04	0.25	0.90	0.27	ε^{Y*}	0.54	0.02	2.55	0.25	0.84	0.395
	68.59	32.61	22.56	26.61	37.59	31.11		52.37	46.06	64.95	61.1	56.12	54.75

Marginal likelihood 3407.80

(a) Log-linear detrending

Marginal likelihood

(b) Log-quadratic detrending

Note: The panels report the forecast error variance decomposition (FEVD) of real income calculated at the posterior mean for the baseline model estimated with alternative filters. Panel (a): Log-linear detrending; Panel (b): Log-quadratic detrending. Marginal Likelihood are computed with Geweke's modified harmonic mean.

7. Extension using Alternative Dynamic Factors

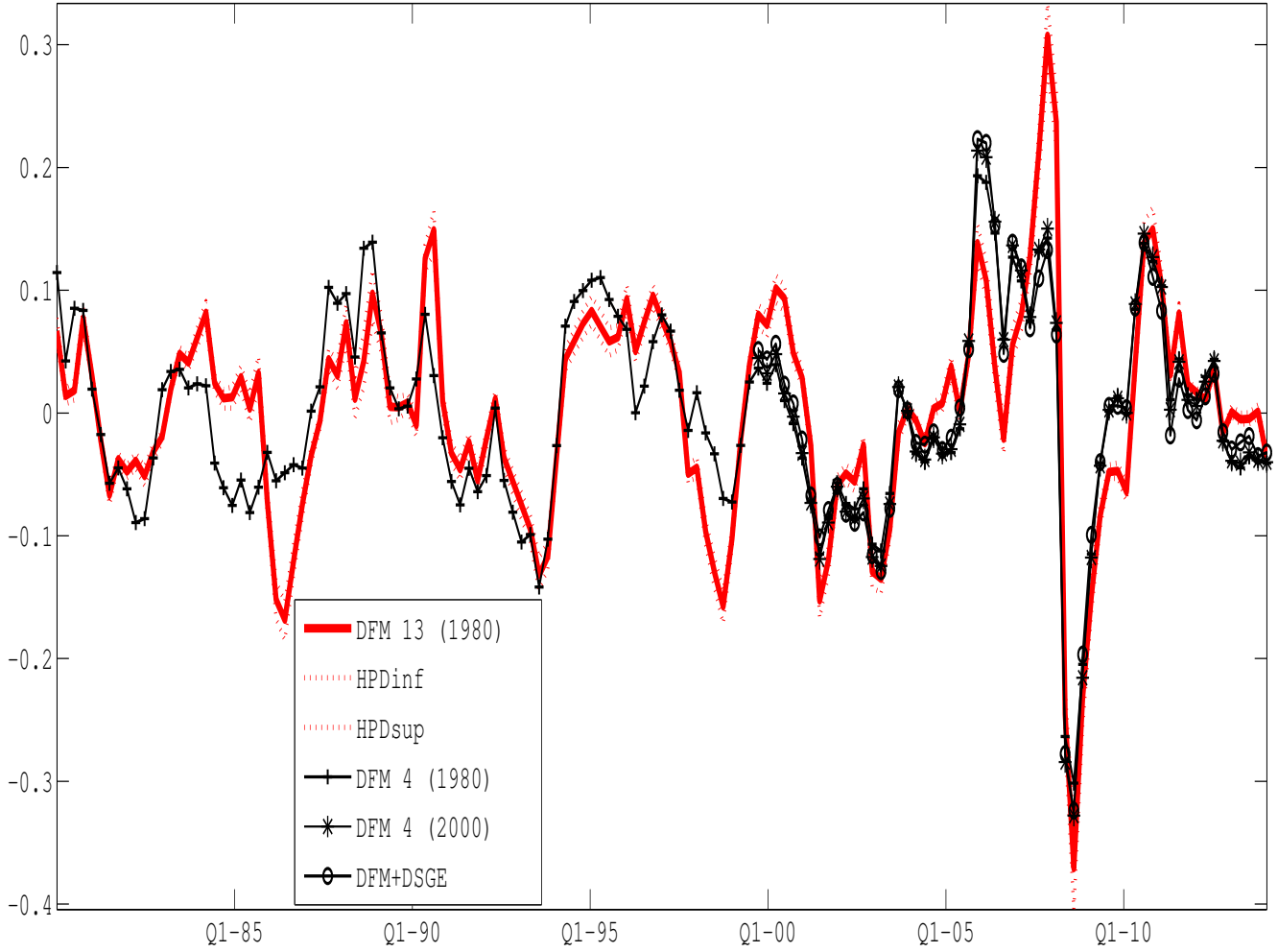
A choice made in the estimation strategy of the benchmark model is the inclusion of the parameters of the dynamic factor structure $\{\phi_{Co}, \rho_j^{Co}, \sigma_j^{Co}, \omega_j^{Co}\}$ in the full set of parameters from the DSGE model that are estimated using full-information methods. As argued in the main text, this choice allows the structural model to identify country-specific, commodity-related, revenue shocks via the idiosyncratic component of the commodity process. This choice, however, entails a tradeoff as it constrains the information on commodity prices used to only the sample period where the DSGE model is estimated, after the year 2000. One might also wonder how the estimation of the dynamic factor model would be if it were done outside the rigid structure imposed by the DSGE on the data.

In this subsection (not reported in the main text) we undertake several extensions that relax this initial choice. First, we estimate the dynamic factor model (DFM) structure in isolation from the structural model, using only data from country-specific commodity price indices. This allows us to use data on commodity prices that goes as far back as 1980 when our time series on single commodity goods prices begin. For this case we also consider using the full range of 13 EMEs in our empirical analysis. We also study the case of only the 4 countries studied in the DSGE model, starting the estimation in 1980 and 2000.⁶

The estimated common factor for each of these alternative cases, together with the benchmark case, that we reproduce for comparison, are reported in Figure A.13. The most noticeable result that can be drawn from this figure is that all robustness cases track down pretty well the common factor estimated in our benchmark estimation. Another noticeable result is that the volatility exhibited in the post-2005 subperiod continues to stand out even when one looks at a longer historical period.

⁶For consistency, the country-specific commodity price indices that start in 1980 continue to use the same commodity weights as those used in the benchmark case. Given that the annual weights do not systematically change much across years for most of the countries considered we think this is an innocuous assumption.

Figure A13. Extended Dynamic Factor Models



Note: The solid/red line is the common factor obtained with a dynamic factor model (DFM) that uses only quarterly data on country-specific commodity prices between 1980.Q1 and 2014.Q3 for the 13 EMEs listed in the text. Dashed/red lines are highest posterior density estimates of confidence bands. DFM 4 (1980) and DFM 4 (2000) reproduce the common factors of two separate DFMs ran with LAC4 countries (Brazil, Chile, Colombia, and Peru) starting in 1980.Q1 and 2000.Q1, respectively. The latter is the starting point of the DSGE estimation. Lastly, DFM+DSGE reproduces the common factor from our benchmark estimation from the DSGE model presented in Figure 5. The common factor is the latent variable obtained from the Kalman filter smoothing evaluated at the mean of the posterior distribution.

8. Extension Assuming Non-Stationarity Commodity Prices

a. Motivation

A hypothesis that has often been tested in the commodities' literature is the presence of non-stationarity in the time series data of commodity prices. While the profession does not seem to have reached a consensus, the evidence appears to favor the rejection of such hypothesis at least when one accounts for the presence of structural breaks (see Mariscal and Powell, 2014, for a recent discussion and evidence). Still, in this robustness section (not reported in the main text) we modify the structural model in order to allow for the possibility that commodity prices follow a unit root process. This can potentially have important consequences insofar as the nature of the commodity price shock can affect the behavior of agents and hence the propagation mechanism by which it affects business cycles.

Formally, we assume that the (now non-stationary) country-specific commodity price index of country j , $P_{j,t}^{Co}$ evolves as follows:

$$P_{t,j}^{Co} = P_{t-1,j}^{Co} g_{j,t}^{Co} = \prod_{s=0}^t g_{j,s}^{Co}$$

where $g_{j,t}^{Co}$ is the (gross) growth rate of the commodity price assumed to follow a dynamic factor structure:

$$\begin{aligned} g_{j,t}^{Co} &= \bar{g}_j + \omega_j^{Co} f_t^{Co} + \varepsilon_{j,t}^{Co} \\ f_t^{Co} &= \phi_{Co} f_{t-1}^{Co} + \sigma^{f^{Co}} \varepsilon_t^{f^{Co}}, \quad \varepsilon_t^{f^{Co}} \sim N(0, 1) \\ \varepsilon_{j,t}^{Co} &= \rho^{Co} \varepsilon_{j,t-1}^{Co} + \sigma_j^{Co} \nu_{j,t}^{Co}, \quad \nu_{j,t}^{Co} \sim N(0, 1) \end{aligned}$$

where $\bar{g}_j$ is assumed to be the long-run (constant, gross) growth rate of $P_{t,j}^{Co}$. Thus, commodity prices' growth rate will fluctuate around this long run mean but transitory shocks to this growth rate, coming either from the common factor or the idiosyncratic component, will have permanent effects on the commodity price level, thereby inducing a non-stationary behavior. The subsection below describes the full-fledged modified model along with the new measurement equations. When taking the model to the data we continue to use growth rates as in the previous subsection.

The results of this extension in terms of priors/posteriors and forecast error variance decomposition of real income are reported in Tables A.28 and A.29, respectively. Overall, the results are consistent with those in the baseline case in terms of the preponderance of commodity price shocks. If anything, the results change in the direction of assigning an even higher share of real income's variance to commodity shocks. The mean/median share of the two commodity price shocks across the four countries considered is 67/87 percent, relative to 47/42 percent in the baseline. In three of the four countries analyzed the share linked to commodity prices is increased (the exception being Brazil). The performance of this extension relative to that of the baseline model in growth rates is considerably worsened as measured by the marginal likelihoods of these two extensions.

b. Summary of Model Equations

$$\begin{aligned} p_t^c \tilde{C}_t + p_t^x \tilde{X}_t + R_{t-1} \frac{\tilde{D}_{t-1}}{g_{t-1}^{PCo}} &= \tilde{w}_t L_t + r_t^k \frac{\tilde{K}_{t-1}}{g_{t-1}^{PCo}} + \tilde{D}_{t-1} + g_t^{PCo} \tilde{C}_o \\ \lambda_t p_t^c &= \left(\tilde{C}_t - \psi^c \frac{L_t^{1+\gamma_C}}{1+\gamma_C} \right)^{-\sigma} \\ \tilde{w}_t &= p_t^c \psi^c L_t^{\gamma_C} \\ \lambda_t &= \left(g_t^{PCo} \right)^{-\sigma} R_t \beta E_t \lambda_{t+1} \\ Q_t^X \lambda_t &= \left(g_t^{PCo} \right)^{-\sigma} \beta E_t \lambda_{t+1} \left[r_{t+1}^k + Q_{t+1}^X (1 - \delta) \right] \\ p_t^x &= Q_t^X \left\{ \begin{aligned} &1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} - 2 \right) \\ &- \frac{\sqrt{a}}{2} \frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} \left[e^{\left(\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} - e^{\left(-\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} \right] \end{aligned} \right\} \\ &+ \left(g_t^{PCo} \right)^{-\sigma} \beta E_t Q_{t+1}^X \frac{\lambda_{t+1}}{\lambda_t} \frac{\sqrt{a}}{2} \left(\frac{\tilde{X}_{t+1}}{\tilde{X}_t} g_t^{PCo} \right)^2 \left(e^{\left(\sqrt{a} \left(\frac{\tilde{X}_{t+1}}{\tilde{X}_t} g_t^{PCo} - \bar{g} \right) \right)} - e^{\left(-\sqrt{a} \left(\frac{\tilde{X}_{t+1}}{\tilde{X}_t} g_t^{PCo} - \bar{g} \right) \right)} \right) \\ \tilde{K}_t &= (1 - \delta) \frac{\tilde{K}_{t-1}}{g_{t-1}^{PCo}} + \tilde{X}_t \left[1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} - 2 \right) \right] \\ \tilde{C}_t &= \left[(1 - \alpha_c)^{\frac{1}{\eta_c}} \left(\tilde{C}_t^h \right)^{\frac{\eta_c - 1}{\eta_c}} + \alpha_c^{\frac{1}{\eta_c}} \left(\tilde{C}_t^f \right)^{\frac{\eta_c - 1}{\eta_c}} \right]^{\frac{\eta_c}{\eta_c - 1}} \\ \tilde{C}_t^h &= (1 - \alpha_c) \left(\frac{p_t^h}{p_t^c} \right)^{-\eta_c} \tilde{C}_t \\ \tilde{C}_t^f &= \alpha_c \left(\frac{1}{p_t^c} \right)^{-\eta_c} \tilde{C}_t \\ \tilde{Y}_t &= z_t \left(\frac{\tilde{K}_{t-1}}{g_{t-1}^{PCo}} \right)^\alpha L_t^{1-\alpha} \\ \tilde{w}_t &= (1 - \alpha) p_t^h \frac{\tilde{Y}_t}{L_t} \end{aligned}$$

$$\begin{aligned}
r_t^k &= \alpha p_t^h \frac{\tilde{Y}_t}{\tilde{K}_{t-1}} g_{t-1}^{PCo} \\
\tilde{X}_t &= \left[(1 - \alpha_x)^{\frac{1}{\eta_x}} \left(\tilde{X}_t^h \right)^{\frac{\eta_x - 1}{\eta_x}} + \alpha_x^{\frac{1}{\eta_x}} \left(\tilde{X}_t^f \right)^{\frac{\eta_x - 1}{\eta_x}} \right]^{\frac{\eta_x}{\eta_x - 1}} \\
\tilde{X}_t^h &= (1 - \alpha_x) \left(\frac{p_t^h}{p_t^x} \right)^{-\eta_x} \tilde{X}_t \\
\tilde{X}_t^f &= \alpha_x \left(\frac{1}{p_t^c} \right)^{-\eta_x} \tilde{X}_t \\
\tilde{Y}_t &= \tilde{C}_t^h + \tilde{X}_t^h + \tilde{C}_t^{h*} \\
G\tilde{D}P_t &= p_t^h \tilde{Y}_t + g_t^{PCo} \overline{Co} \\
RER_t &= \frac{1}{p_t^c} \\
TBY_t &= \frac{p_t^h \tilde{C}_t^{h*} + g_t^{PCo} \overline{Co} - \tilde{C}_t^f - \tilde{X}_t^f}{\tilde{Y}_t} \\
\tilde{C}_{j,t}^{h*} &= \left(p_{j,t}^h \right)^{-\epsilon_{j,e}} \tilde{Y}_t^*
\end{aligned}$$

$$\begin{aligned}
R_{jt} &= R_t^* S_{jt} \\
S_{jt} &= z_{jt}^R \left\{ \exp \left[\Omega_{j,u} (\tilde{D}_{j,t} - \overline{D}_j) \right] - 1 \right\}
\end{aligned}$$

$$\begin{aligned}
g_{j,t}^{Co} &= \overline{g}_j + \omega_j^{Co} f_t^{Co} + \varepsilon_{j,t}^{Co} \\
f_t^{Co} &= \phi_{Co} f_{t-1}^{Co} + \sigma^{f^{Co}} \varepsilon_t^{f^{Co}} \\
\varepsilon_{j,t}^{Co} &= \rho^{Co} \varepsilon_{j,t-1}^{Co} + \sigma_j^{Co} \nu_{j,t}^{Co}
\end{aligned}$$

$$\begin{aligned}
\ln R_t^* &= \rho_{R^*} \ln R_{t-1}^* + (1 - \rho_{R^*}) \ln \overline{R}^* + \sigma^{R^*} \varepsilon_t^{R^*} \\
z_{j,t}^R &= \left(1 - \rho_j^R \right) \overline{z}_j^R + \rho_j^R z_{j,t-1}^R + \sigma_j^{z^R} \nu_{j,t}^R \\
\ln \tilde{Y}_t^* &= (1 - \rho_{Y^*}) \ln \overline{Y}^* + \rho_{Y^*} \ln \tilde{Y}_{t-1}^* + \sigma^{Y^*} \varepsilon_t^{Y^*} \\
\ln z_{j,t} &= (1 - \rho_j^z) \ln \tilde{z}_j + \rho_j^z \ln z_{j,t-1} + \sigma_j^{z^z} \varepsilon_{j,t}^{z^z}
\end{aligned}$$

c. Measurement Equations

- Preferences

$$\begin{aligned}
[Benchmark] : u(C_t, L_t) &= \frac{1}{1 - \sigma} \left(C_t - \psi^c \frac{L_t^{1+\gamma_C}}{1 + \gamma_C} \right)^{1-\sigma} \\
[Robustness] : u(C_t, L_t) &= \frac{1}{1 - \sigma} \left(C_t - p_{t-1}^{Co} \psi^c \frac{L_t^{1+\gamma_C}}{1 + \gamma_C} \right)^{1-\sigma}
\end{aligned}$$

- New law of motion for p_t^{Co} which allows us to model the nonstationarity of this price:

$$p_t^{Co} = p_{t-1}^{Co} g_t^{PCo}$$

- Resource constraint

$$\begin{aligned}
[Benchmark] : p_t^c C_t + p_t^x X_t + R_{t-1} D_{t-1} &= w_t L_t + r_t^k K_{t-1} + D_t + p_t^{Co} \overline{Co} \\
[Robustness] : p_t^c \frac{C_t}{p_{t-1}^{Co}} + p_t^x \frac{X_t}{p_{t-1}^{Co}} + R_{t-1} \frac{D_{t-1}}{p_{t-2}^{Co} p_{t-1}^{Co}} &= \frac{w_t}{p_{t-1}^{Co}} L_t + r_t^k \frac{K_{t-1}}{p_{t-2}^{Co} p_{t-1}^{Co}} + \frac{D_t}{p_{t-1}^{Co}} + \frac{p_t^{Co}}{p_{t-1}^{Co}} \overline{Co} \\
[Robustness] : p_t^c \tilde{C}_t + p_t^x \tilde{X}_t + R_{t-1} \frac{\tilde{D}_{t-1}}{g_{t-1}^{PCo}} &= \tilde{w}_t L_t + r_t^k \frac{\tilde{K}_{t-1}}{g_{t-1}^{PCo}} + \tilde{D}_{t-1} + g_t^{PCo} \overline{Co}
\end{aligned}$$

where

$$\tilde{C}_t = \frac{C_t}{p_{t-1}^{Co}}, \quad \tilde{X}_t = \frac{X_t}{p_{t-1}^{Co}}, \quad \tilde{D}_t = \frac{D_t}{p_{t-1}^{Co}}, \quad \tilde{w}_t = \frac{w_t}{p_{t-1}^{Co}}, \quad \tilde{K}_t = \frac{K_t}{p_{t-1}^{Co}}$$

- Capital law of motion

$$\begin{aligned}
[Benchmark] : K_t &= (1 - \delta) K_{t-1} + X_t \left[1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} - 2 \right) \right] \\
[Robustness] : \frac{K_t}{p_{t-1}^{Co}} &= (1 - \delta) \frac{K_{t-1}}{p_{t-2}^{Co}} \frac{p_{t-2}^{Co}}{p_{t-1}^{Co}} \\
&+ \frac{X_t}{p_{t-1}^{Co}} \left[1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} \frac{p_{t-1}^{Co}}{p_{t-1}^{Co}} \frac{p_{t-2}^{Co}}{p_{t-2}^{Co}} - \bar{g} \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} \frac{p_{t-1}^{Co}}{p_{t-1}^{Co}} \frac{p_{t-2}^{Co}}{p_{t-2}^{Co}} - \bar{g} \right) \right)} - 2 \right) \right] \\
[Robustness] : \tilde{K}_t &= (1 - \delta) \frac{\tilde{K}_{t-1}}{g_{t-1}^{PCo}} + \tilde{X}_t \left[1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} - 2 \right) \right]
\end{aligned}$$

- Optimization problem:

$$\begin{aligned}
&\max_{\{C_t, L_t, X_t, D_t\}} E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, L_t) \\
&s.t \\
&(i) \text{ Resource constraint} \\
&(ii) \text{ Capital law of motion}
\end{aligned}$$

$$\begin{aligned}
[Benchmark] : \mathcal{L} &= E_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{1 - \sigma} \left(C_t - \psi^c \frac{L_t^{1+\gamma_C}}{1 + \gamma_C} \right)^{1-\sigma} \\
&+ E_0 \sum_{t=0}^{\infty} \lambda_t \beta^t \left(w_t L_t + r_t^k K_{t-1} + D_t + p_t^{Co} \bar{C}o - p_t^c C_t - p_t^x X_t - R_{t-1} D_{t-1} \right) \\
&+ E_0 \sum_{t=0}^{\infty} Q_t^X \lambda_t \beta^t \left\{ +X_t \left[1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} - 2 \right) \right] - K_t \right\}
\end{aligned}$$

FOC :

$$\begin{aligned}
[C_t] : \beta^t \left(C_t - \psi^c \frac{L_t^{1+\gamma_C}}{1 + \gamma_C} \right)^{-\sigma} + \beta^t \lambda_t (-p_t^c) &= 0 \\
[L_t] : \beta^t \left(C_t - \psi^c \frac{L_t^{1+\gamma_C}}{1 + \gamma_C} \right)^{-\sigma} (-\psi^c L_t^{\gamma_C}) + \beta^t \lambda_t (w_t) &= 0 \\
[D_t] : \lambda_t \beta^t - R_t \beta^{t+1} E_t \lambda_{t+1} &= 0 \\
[K_t] : Q_t^X \lambda_t \beta^t \{-1\} + \beta^{t+1} E_t \lambda_{t+1} r_{t+1}^k + \beta^{t+1} E_t Q_{t+1}^X \lambda_{t+1} (1 - \delta) &= 0 \\
[X_t] : \lambda_t \beta^t (-p_t^x) + \\
&Q_t^X \lambda_t \beta^t \left\{ \begin{aligned} &\left[1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} - 2 \right) \right] \\ &+ X_t \left[-\frac{1}{2} \left(\left(\frac{\sqrt{a}}{X_{t-1}} \right) e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} + \left(\frac{-\sqrt{a}}{X_{t-1}} \right) e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} \right) \right] \end{aligned} \right\} \\
&- E_t Q_{t+1}^X \lambda_{t+1} \beta^{t+1} \left\{ X_{t+1} \frac{1}{2} \left(\begin{aligned} &\left(-\sqrt{a} \frac{X_{t+1}}{X_t^2} \right) e^{\left(\sqrt{a} \left(\frac{X_{t+1}}{X_t} - 1 \right) \right)} \\ &+ \left(\sqrt{a} \frac{X_{t+1}}{X_t^2} \right) e^{\left(-\sqrt{a} \left(\frac{X_{t+1}}{X_t} - 1 \right) \right)} \end{aligned} \right) \right\} \\
&= 0
\end{aligned}$$

Which gives rise to these more compact FOC:

$$\begin{aligned}
[Benchmark] : \lambda_t p_t^c &= \left(C_t - \psi^c \frac{L_t^{1+\gamma_C}}{1 + \gamma_C} \right)^{-\sigma} \\
[Benchmark] : w_t &= p_t^c \psi^c L_t^{\gamma_C} \\
[Benchmark] : \lambda_t &= R_t \beta E_t \lambda_{t+1} \\
[Benchmark] : Q_t^X \lambda_t &= \beta E_t \lambda_{t+1} \left[r_{t+1}^k + Q_{t+1}^X (1 - \delta) \right]
\end{aligned}$$

$$[Benchmark] : p_t^x = Q_t^X \left\{ 1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} - 2 \right) - \frac{\sqrt{a}}{2} \frac{X_t}{X_{t-1}} \left[e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} - e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} \right] \right\} \\ + \beta E_t Q_{t+1}^X \frac{\lambda_{t+1}}{\lambda_t} \frac{\sqrt{a}}{2} \left(\frac{X_{t+1}}{X_t} \right)^2 \left(e^{\left(\sqrt{a} \left(\frac{X_{t+1}}{X_t} - 1 \right) \right)} - e^{\left(-\sqrt{a} \left(\frac{X_{t+1}}{X_t} - 1 \right) \right)} \right)$$

This changes as follows in the alternative case

$$[Robustness] : \mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{1-\sigma} \left(C_t - p_{t-1}^{Co} \psi^c \frac{L_t^{1+\gamma_C}}{1+\gamma_C} \right)^{1-\sigma} \\ + E_0 \sum_{t=0}^{\infty} \lambda_t \beta^t \left(p_{t-1}^{Co} \right)^{-\sigma} \left(w_t L_t + r_t^k K_{t-1} + D_t + p_t^{Co} \bar{C}o - p_t^c C_t - p_t^x X_t - R_{t-1} D_{t-1} \right) \\ + E_0 \sum_{t=0}^{\infty} Q_t^X \lambda_t \beta^t \left(p_{t-1}^{Co} \right)^{-\sigma} \left\{ + X_t \left[1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} - 2 \right) \right] - K_t \right\}$$

FOC :

$$[C_t] : \beta^t \left(C_t - p_{t-1}^{Co} \psi^c \frac{L_t^{1+\gamma_C}}{1+\gamma_C} \right)^{-\sigma} + \beta^t \lambda_t \left(p_{t-1}^{Co} \right)^{-\sigma} (-p_t^c) = 0 \\ [L_t] : \beta^t \left(C_t - p_{t-1}^{Co} \psi^c \frac{L_t^{1+\gamma_C}}{1+\gamma_C} \right)^{-\sigma} \left(-p_{t-1}^{Co} \psi^c L_t^{\gamma_C} \right) + \beta^t \lambda_t \left(p_{t-1}^{Co} \right)^{-\sigma} (w_t) = 0 \\ [D_t] : \lambda_t \beta^t \left(p_{t-1}^{Co} \right)^{-\sigma} - \left(p_t^{Co} \right)^{-\sigma} R_t \beta^{t+1} E_t \lambda_{t+1} = 0 \\ [K_t] : Q_t^X \lambda_t \beta^t \left(p_{t-1}^{Co} \right)^{-\sigma} \{-1\} + \beta^{t+1} \left(p_t^{Co} \right)^{-\sigma} E_t \lambda_{t+1} r_{t+1}^k + \beta^{t+1} \left(p_t^{Co} \right)^{-\sigma} E_t Q_{t+1}^X \lambda_{t+1} (1-\delta) = 0 \\ [X_t] : \lambda_t \beta^t \left(p_{t-1}^{Co} \right)^{-\sigma} (-p_t^x) \\ + Q_t^X \lambda_t \beta^t \left(p_{t-1}^{Co} \right)^{-\sigma} \left\{ \left[1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} - 2 \right) \right] \right. \\ \left. + X_t \left[-\frac{1}{2} \left(\left(\frac{\sqrt{a}}{X_{t-1}} \right) e^{\left(\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} + \left(\frac{-\sqrt{a}}{X_{t-1}} \right) e^{\left(-\sqrt{a} \left(\frac{X_t}{X_{t-1}} - 1 \right) \right)} \right) \right] \right\} \\ - E_t Q_{t+1}^X \lambda_{t+1} \beta^{t+1} \left(p_t^{Co} \right)^{-\sigma} \left\{ X_{t+1} \frac{1}{2} \left(\begin{array}{l} \left(-\sqrt{a} \frac{X_{t+1}}{X_t^2} \right) e^{\left(\sqrt{a} \left(\frac{X_{t+1}}{X_t} - 1 \right) \right)} \\ + \left(\sqrt{a} \frac{X_{t+1}}{X_t^2} \right) e^{\left(-\sqrt{a} \left(\frac{X_{t+1}}{X_t} - 1 \right) \right)} \end{array} \right) \right\} \\ = 0$$

Which gives rise to these more compact FOC:

$$[Robustness] : p_t^x = Q_t^X \left\{ \begin{array}{l} 1 - \frac{1}{2} \left(e^{\left(\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} + e^{\left(-\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} - 2 \right) \\ - \frac{\sqrt{a}}{2} \frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} \left[e^{\left(\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} - e^{\left(-\sqrt{a} \left(\frac{\tilde{X}_t}{\tilde{X}_{t-1}} g_{t-1}^{PCo} - \bar{g} \right) \right)} \right] \end{array} \right\} \\ + \left(g_t^{PCo} \right)^{-\sigma} \beta E_t Q_{t+1}^X \frac{\lambda_{t+1}}{\lambda_t} \frac{\sqrt{a}}{2} \left(\frac{\tilde{X}_{t+1}}{\tilde{X}_t} g_t^{PCo} \right)^2 \left(e^{\left(\sqrt{a} \left(\frac{\tilde{X}_{t+1}}{\tilde{X}_t} g_t^{PCo} - \bar{g} \right) \right)} - e^{\left(-\sqrt{a} \left(\frac{\tilde{X}_{t+1}}{\tilde{X}_t} g_t^{PCo} - \bar{g} \right) \right)} \right) \\ [Robustness] : Q_t^X \lambda_t = \left(g_t^{PCo} \right)^{-\sigma} \beta E_t \lambda_{t+1} \left[r_{t+1}^k + Q_{t+1}^X (1-\delta) \right] \\ [Robustness] : \lambda_t = \left(g_t^{PCo} \right)^{-\sigma} R_t \beta E_t \lambda_{t+1} \\ [Robustness] : \tilde{w}_t = p_t^c \psi^c L_t^{\gamma_C} \\ [Robustness] : \lambda_t p_t^c = \left(\tilde{C}_t - \psi^c \frac{L_t^{1+\gamma_C}}{1+\gamma_C} \right)^{-\sigma}$$

- Agg. Consumption Bundle

$$[Benchmark] : C_t = \left[(1-\alpha_c)^{\frac{1}{\eta_c}} \left(C_t^h \right)^{\frac{\eta_c-1}{\eta_c}} + \alpha_c^{\frac{1}{\eta_c}} \left(C_t^f \right)^{\frac{\eta_c-1}{\eta_c}} \right]^{\frac{\eta_c}{\eta_c-1}} \\ [Robustness] : \tilde{C}_t = \left[(1-\alpha_c)^{\frac{1}{\eta_c}} \left(\tilde{C}_t^h \right)^{\frac{\eta_c-1}{\eta_c}} + \alpha_c^{\frac{1}{\eta_c}} \left(\tilde{C}_t^f \right)^{\frac{\eta_c-1}{\eta_c}} \right]^{\frac{\eta_c}{\eta_c-1}}$$

- Optimal Eqs. for h- and f- consumption

$$[Benchmark] : C_t^h = (1 - \alpha_c) \left(\frac{p_t^h}{p_t^c} \right)^{-\eta_c} C_t$$

$$[Robustness] : \tilde{C}_t^h = (1 - \alpha_c) \left(\frac{p_t^h}{p_t^c} \right)^{-\eta_c} \tilde{C}_t$$

$$[Benchmark] : C_t^f = \alpha_c \left(\frac{1}{p_t^c} \right)^{-\eta_c} C_t$$

$$[Robustness] : \tilde{C}_t^f = \alpha_c \left(\frac{1}{p_t^c} \right)^{-\eta_c} \tilde{C}_t$$

- Production Function

$$[Benchmark] : Y_t = z_t K_{t-1}^\alpha \left(p_t^{Co} L_t \right)^{1-\alpha}$$

$$[Robustness] : \frac{Y_t}{p_{t-1}^{Co}} = z_t \left(\frac{K_{t-1} p_{t-2}^{Co}}{p_{t-1}^{Co} p_{t-2}^{Co}} \right)^\alpha \left(\frac{p_t^{Co}}{p_{t-1}^{Co}} L_t \right)^{1-\alpha}$$

$$[Robustness] : \tilde{Y}_t = z_t \left(\frac{\tilde{K}_{t-1}}{g_{t-1}^{PCo}} \right)^\alpha L_t^{1-\alpha}$$

- Production of h-goods

$$[Benchmark] : \max_{\{K_{t-1}, L_t\}} \xi_t = p_t^h z_t K_{t-1}^\alpha L_t^{1-\alpha} - w_t L_t - r_t^k K_{t-1}$$

$$[K_{t-1}] : \alpha p_t^h z_t K_{t-1}^{\alpha-1} L_t^{1-\alpha} - r_t^k = 0$$

$$\rightarrow r_t^k = \alpha p_t^h \frac{z_t K_{t-1}^\alpha L_t^{1-\alpha}}{K_{t-1}} = \alpha p_t^h \frac{Y_t}{K_{t-1}}$$

$$[w_t] : (1 - \alpha) p_t^h z_t K_{t-1}^\alpha L_t^{-\alpha} - w_t = 0$$

$$\rightarrow w_t = (1 - \alpha) p_t^h \frac{z_t K_{t-1}^\alpha L_t^{1-\alpha}}{L_t} = (1 - \alpha) p_t^h \frac{Y_t}{L_t}$$

$$[Robustness] : \max_{\{K_{t-1}, L_t\}} \xi_t = p_t^h z_t K_{t-1}^\alpha \left(p_t^{Co} L_t \right)^{1-\alpha} - w_t L_t - r_t^k K_{t-1}$$

$$[K_{t-1}] : \alpha p_t^h z_t K_{t-1}^{\alpha-1} \left(p_t^{Co} L_t \right)^{1-\alpha} - r_t^k = 0$$

$$\rightarrow r_t^k = \alpha p_t^h \frac{z_t K_{t-1}^\alpha \left(p_t^{Co} L_t \right)^{1-\alpha}}{K_{t-1}}$$

$$\rightarrow r_t^k = \alpha p_t^h \frac{\tilde{Y}_t}{\tilde{K}_{t-1}} g_{t-1}^{PCo}$$

$$[w_t] : (1 - \alpha) p_t^h z_t K_{t-1}^\alpha \left(p_t^{Co} L_t \right)^{-\alpha} p_t^{Co} - w_t = 0$$

$$\rightarrow w_t = (1 - \alpha) p_t^h \frac{z_t K_{t-1}^\alpha \left(p_t^{Co} L_t \right)^{1-\alpha}}{L_t} = (1 - \alpha) p_t^h \frac{Y_t}{L_t}$$

$$\rightarrow \tilde{w}_t = (1 - \alpha) p_t^h \frac{\tilde{Y}_t}{L_t}$$

- Agg. Investment Bundle

$$[Benchmark] : X_t = \left[(1 - \alpha_x)^{\frac{1}{\eta_x}} \left(X_t^h \right)^{\frac{\eta_x - 1}{\eta_x}} + \alpha_x^{\frac{1}{\eta_x}} \left(X_t^f \right)^{\frac{\eta_x - 1}{\eta_x}} \right]^{\frac{\eta_x}{\eta_x - 1}}$$

$$[Robustness] : \tilde{X}_t = \left[(1 - \alpha_x)^{\frac{1}{\eta_x}} \left(\tilde{X}_t^h \right)^{\frac{\eta_x - 1}{\eta_x}} + \alpha_x^{\frac{1}{\eta_x}} \left(\tilde{X}_t^f \right)^{\frac{\eta_x - 1}{\eta_x}} \right]^{\frac{\eta_x}{\eta_x - 1}}$$

- Optimal Eqs. for h- and f- investment

$$[Benchmark] : X_t^h = (1 - \alpha_x) \left(\frac{p_t^h}{p_t^x} \right)^{-\eta_x} X_t$$

$$[Robustness] : \tilde{X}_t^h = (1 - \alpha_x) \left(\frac{p_t^h}{p_t^x} \right)^{-\eta_x} \tilde{X}_t$$

$$[Benchmark] : X_t^f = \alpha_x \left(\frac{1}{p_t^c} \right)^{-\eta_x} X_t$$

$$[Robustness] : \tilde{X}_t^f = \alpha_x \left(\frac{1}{p_t^c} \right)^{-\eta_x} \tilde{X}_t$$

- Market Eq.

$$[Benchmark] : Y_t = C_t^h + X_t^h + C_t^{h*}$$

$$[Robustness] : \tilde{Y}_t = \tilde{C}_t^h + \tilde{X}_t^h + \tilde{C}_t^{h*}$$

- GDP Def.

$$[Benchmark] : GDP_t = p_t^h Y_t + p_t^{Co} \overline{Co}$$

$$[Robustness] : G\tilde{D}P_t = p_t^h \tilde{Y}_t + g_t^{PCo} \overline{Co}$$

- RER Def.

$$[Benchmark] : RER_t = \frac{1}{p_t^c}$$

$$[Robustness] : RER_t = \frac{1}{p_t^c}$$

- Trade Balance share

$$[Benchmark] : TBY_t = \frac{p_t^h C_t^{h*} + p_t^{Co} \overline{Co} - C_t^f - X_t^f}{Y_t}$$

$$[Robustness] : TBY_t = \frac{p_t^h \tilde{C}_t^{h*} + g_t^{PCo} \overline{Co} - \tilde{C}_t^f - \tilde{X}_t^f}{\tilde{Y}_t}$$

- h -good external demand

$$[Benchmark] : C_{j,t}^{h*} = \left(p_{j,t}^h \right)^{-\epsilon_{j,e}} Y_t^*$$

$$[Robustness] : \tilde{C}_{j,t}^{h*} = \left(p_{j,t}^h \right)^{-\epsilon_{j,e}} \tilde{Y}_t^*$$

- Interest Rate and Spreads

$$[Benchmark] : R_{jt} = R_t^* S_{jt}$$

$$[Benchmark] : S_{jt} = z_{jt}^R \left\{ \exp \left[\Omega_{j,u} (D_{j,t} - \bar{D}_j) \right] - 1 \right\}$$

$$[Robustness] : R_{jt} = R_t^* S_{jt}$$

$$[Robustness] : S_{jt} = z_{jt}^R \left\{ \exp \left[\Omega_{j,u} (\tilde{D}_{j,t} - \bar{D}_j) \right] - 1 \right\}$$

- DFM

$$[Benchmark] : p_{j,t}^{Co} = \omega_j^{Co} f_t^{Co} + \varepsilon_{j,t}^{Co}$$

$$[Benchmark] : f_t^{Co} = \phi_{Co} f_{t-1}^{Co} + \sigma^{f^{Co}} \varepsilon_t^{f^{Co}}$$

$$[Benchmark] : \varepsilon_{j,t}^{Co} = \rho^{Co} \varepsilon_{j,t-1}^{Co} + \sigma_j^{Co} \nu_{j,t}^{Co}$$

$$[Robustness] : g_{j,t}^{Co} = \bar{g}_j + \omega_j^{Co} f_t^{Co} + \varepsilon_{j,t}^{Co}$$

$$[Robustness] : f_t^{Co} = \phi_{Co} f_{t-1}^{Co} + \sigma^{f^{Co}} \varepsilon_t^{f^{Co}}$$

$$[Robustness] : \varepsilon_{j,t}^{Co} = \rho^{Co} \varepsilon_{j,t-1}^{Co} + \sigma_j^{Co} \nu_{j,t}^{Co}$$

- Remaining Driving Forces

$$\ln R_t^* = \rho_{R^*} \ln R_{t-1}^* + (1 - \rho_{R^*}) \ln \bar{R}^* + \sigma^{R^*} \varepsilon_t^{R^*}$$

$$z_{j,t}^R = \left(1 - \rho_j^{z^R} \right) \bar{z}_j^R + \rho_j^{z^R} z_{j,t-1}^R + \sigma_j^{z^R} \nu_{j,t}^R$$

$$\ln \tilde{Y}_t^* = (1 - \rho_{Y^*}) \ln \bar{Y}^* + \rho_{Y^*} \ln \tilde{Y}_{t-1}^* + \sigma^{Y^*} \varepsilon_t^{Y^*}$$

$$\ln z_{j,t} = (1 - \rho_j^z) \ln \bar{z}_j + \rho_j^z \ln z_{j,t-1} + \sigma_j^z \varepsilon_{j,t}^z$$

d. Priors/Posteriors Results

Table A28. Priors and Posteriors of Estimated Parameters Non Stationary Commodity Prices

Parameters	Type	Prior		Mode	S.D.	Posterior		
		Mean	S.D.			Mean	90 % HPD interval	
Global								
ρ_{Y^*}	beta	0.5	0.15	0.9234	0.0262	0.9031	0.8577	0.9494
ρ_{R^*}	beta	0.5	0.15	0.664	0.0658	0.6452	0.5447	0.7512
ϕ_{Co}	beta	0.5	0.15	0.0127	0.0062	0.0187	0.0062	0.03
σ^{Y^*}	invg	0.013	Inf	0.0067	0.0006	0.0068	0.0058	0.0078
σ^{R^*}	invg	0.013	Inf	0.0028	0.0003	0.0026	0.0022	0.003
σ^{fCo}	invg	0.013	Inf	0.1084	0.0113	0.102	0.0863	0.1174
Brazil								
ρ^{zR}	beta	0.5	0.15	0.2003	0.0634	0.1525	0.0671	0.2369
ρ^{Co}	beta	0.5	0.15	0.0584	0.026	0.0607	0.0212	0.0973
ρ^z	beta	0.5	0.15	0.996	0	0.996	0.996	0.996
a	gamma	0.5	0.25	0.7436	0.2054	1.0603	0.5675	1.5416
ω^{Co}	norm	0	1	0.4731	0.0649	0.4364	0.3349	0.5394
σ^{zR}	invg	0.01	Inf	0.0061	0.0006	0.0063	0.0054	0.0073
σ^{Co}	invg	0.01	Inf	0.0483	0.0046	0.0484	0.0411	0.0554
σ^z	invg	0.01	Inf	0.0338	0.0029	0.035	0.0297	0.0401
Chile								
ρ^{zR}	beta	0.5	0.15	0.5617	0.1142	0.5815	0.398	0.7585
ρ^{Co}	beta	0.5	0.15	0.0725	0.0304	0.0675	0.0251	0.108
ρ^z	beta	0.5	0.15	0.6114	0.0588	0.5849	0.4834	0.6875
a	gamma	0.5	0.25	3.5934	0.6129	2.9903	2.292	3.7148
ω^{Co}	norm	0	1	0.8772	0.1013	0.8151	0.6584	0.9854
σ^{zR}	invg	0.01	Inf	0.0018	0.0001	0.0019	0.0016	0.0022
σ^{Co}	invg	0.01	Inf	0.0742	0.0068	0.0765	0.0654	0.088
σ^z	invg	0.01	Inf	0.0772	0.0075	0.0847	0.0692	0.0993
Colombia								
ρ^{zR}	beta	0.5	0.15	0.235	0.08	0.3257	0.1948	0.4598
ρ^{Co}	beta	0.5	0.15	0.936	0.0221	0.9222	0.8874	0.9561
ρ^z	beta	0.5	0.15	0.8224	0.057	0.8071	0.7187	0.8988
a	gamma	0.5	0.25	1.2823	0.3983	1.0726	0.607	1.5418
σ^{zR}	invg	0.01	Inf	0.0032	0.0003	0.0033	0.0028	0.0038
σ^{Co}	invg	0.01	Inf	0.0077	0.002	0.0105	0.0063	0.0142
σ^z	invg	0.01	Inf	0.0312	0.0031	0.0318	0.0269	0.0367
Peru								
ρ^{zR}	beta	0.5	0.15	0.2526	0.0836	0.2766	0.143	0.4078
ρ^{Co}	beta	0.5	0.15	0.1277	0.0464	0.1419	0.0661	0.2152
ρ^z	beta	0.5	0.15	0.8498	0.0368	0.8594	0.8008	0.9145
a	gamma	0.5	0.25	1.8117	0.4267	1.7344	1.0985	2.3401
ω^{Co}	norm	0	1	0.4383	0.0655	0.4905	0.3773	0.6015
σ^{zR}	invg	0.01	Inf	0.0028	0.0003	0.0026	0.0022	0.0031
σ^{Co}	invg	0.01	Inf	0.0473	0.0041	0.0502	0.0427	0.0575
σ^z	invg	0.01	Inf	0.0149	0.0015	0.0149	0.0125	0.0172

Note: This table shows the priors and posteriors based on 200,000 draws from the Metropolis-Hastings (MH) algorithm, discarding the first 100,000 draws. The mean and covariance matrix of the proposal density for the MH algorithm were the maximum of the posterior distribution and the negative inverse Hessian around that maximum obtained with Nelder-Mead simplex based optimization routine. The computations were conducted using Dynare 4.4.2. HPD stands for higher posterior density.

8. Cyclical Production of Commodities: A Discussion

Arguably the assumption of an endowment flow of commodity goods in the model is not without loss of generality. Ideally a more complete model of an emerging economy would include production in this sector along other features associated to the economics of commodities like storage capacity, optimal management of inventories, foreign ownership of capital, monopolistic rents, speculation, etc.⁷ Yet we believe that such realistic features would likely make the analytical framework less tractable, and would obscure the main propagation channel that we think is the most important from a business cycle perspective: the revenue shock that comes with movements in the world markets of commodity goods.

Nonetheless, we think that further discussion about the validity of this assumption is warranted. Thus, in this subsection we present the available evidence regarding the cyclical production over the business cycle and discuss its implications. Figure A.14, panels (a) to (h), reports the serial correlation between the volumetric production of the two most important commodities in each of the four countries considered in the estimation of the structural model with the international market price of each commodity. The sample period over which the correlations are computed is the same as the one used when estimating the structural model. The results indicate no systematic comovement whatsoever between quantities and prices. Indeed, in most cases, commodity production does not react to changes in prices at business cycle frequencies. Data from the volumetric quantities of each of the two commodities is not reported in UN Comtrade. We use national sources for each commodity. Fig. A.15 reports the same statistics presented in this subsection for the entire period for which data on volumetric quantities is available. Results are qualitatively robust. A more formal analysis with the use of Granger-causality tests between the two variables reveals no causality in either direction (see Table A.30). This not only provides evidence of the absence of a response of commodity production to changes in commodity prices, but also backs the assumption that these countries are price takers in international markets.

The remaining two subplots in Figure A.15, (i) and (j), present further evidence related to the propagation mechanism of the revenue shock implied by the model. Subplot (i) reports the serial correlation between the simple average of the non-commodities GDP for the four countries studied in the structural estimation and their country-specific commodity price indices.⁸ Subplot (j) looks instead at the nontraded GDP and its comovement with commodity prices across countries. Because constraints on data availability are less stringent for non-traded output, the latter is done for a wider pool of EMEs. Regardless of which of the two measures of real economic activity one looks at, there is a considerable comovement between them and country commodity price indices. This goes in line with the main prediction of the model's propagation mechanism following a commodity shock: that economic activity, mainly the one that is unrelated to the commodity sector, comoves with commodity prices, driven by changes in real income which, in turn, come from exogenous fluctuations in those prices.

⁷For a model of commodity price dynamics with some of these features, mainly storage and speculation behavior, see Deaton and Laroque (1996).

⁸Non-commodity production for Brazil is computed as the difference between total GDP and GDP in agriculture and mining. In Chile, it is defined as total GDP excluding copper mining and fishing sectors. Non-commodity production in Colombia is computed as the total GDP minus GDP in coffee and mining (which includes crude oil production). Finally, in Peru it is defined as total GDP excluding mining and hydrocarbon sectors.

e. FEVD Results

Table A29. Forecast Error Variance Decomposition of Output: Non Stationary Commodity Prices

shocks	Brazil	Chile	Colombia	Peru	Mean	Median	shocks	Brazil	Chile	Colombia	Peru	Mean	Median
Commodity price shock							Commodity price shock						
ν^{Co}	16.3	0.5	17.6	4.9	9.8	10.6	ν^{Co}	1.0	33.8	53.9	51.3	35.0	42.5
ε^{fCo}	11.1	76.6	25.8	35.5	37.3	30.7	ε^{fCo}	1.0	49.6	36.2	40.7	31.9	38.4
	27.5	77.1	43.5	40.4	47.1	42.0		2.0	83.4	90.1	92.0	66.9	86.7
All other shocks							All other shocks						
ν^R	30.0	0.03	4.43	0.3	8.7	2.4	ν^R	0.1	0.0	0.0	0.0	0.0	0.0
ε^z	22.0	22.9	45.4	58.3	37.2	34.2	ε^z	97.8	16.6	9.8	8.0	33.1	13.2
ε^{R*}	20.3	0.03	5.8	0.92	6.8	3.4	ε^{R*}	0.1	0.0	0.1	0.0	0.0	0.0
ε^{Y*}	0.19	0.01	0.9	0.1	0.3	0.1	ε^{Y*}	0.0	0.0	0.0	0.0	0.0	0.0
	72.5	22.9	56.5	59.6	52.9	58.1		98.0	16.6	9.9	8.0	33.1	13.3

Marginal likelihood

4530.9

Marginal likelihood

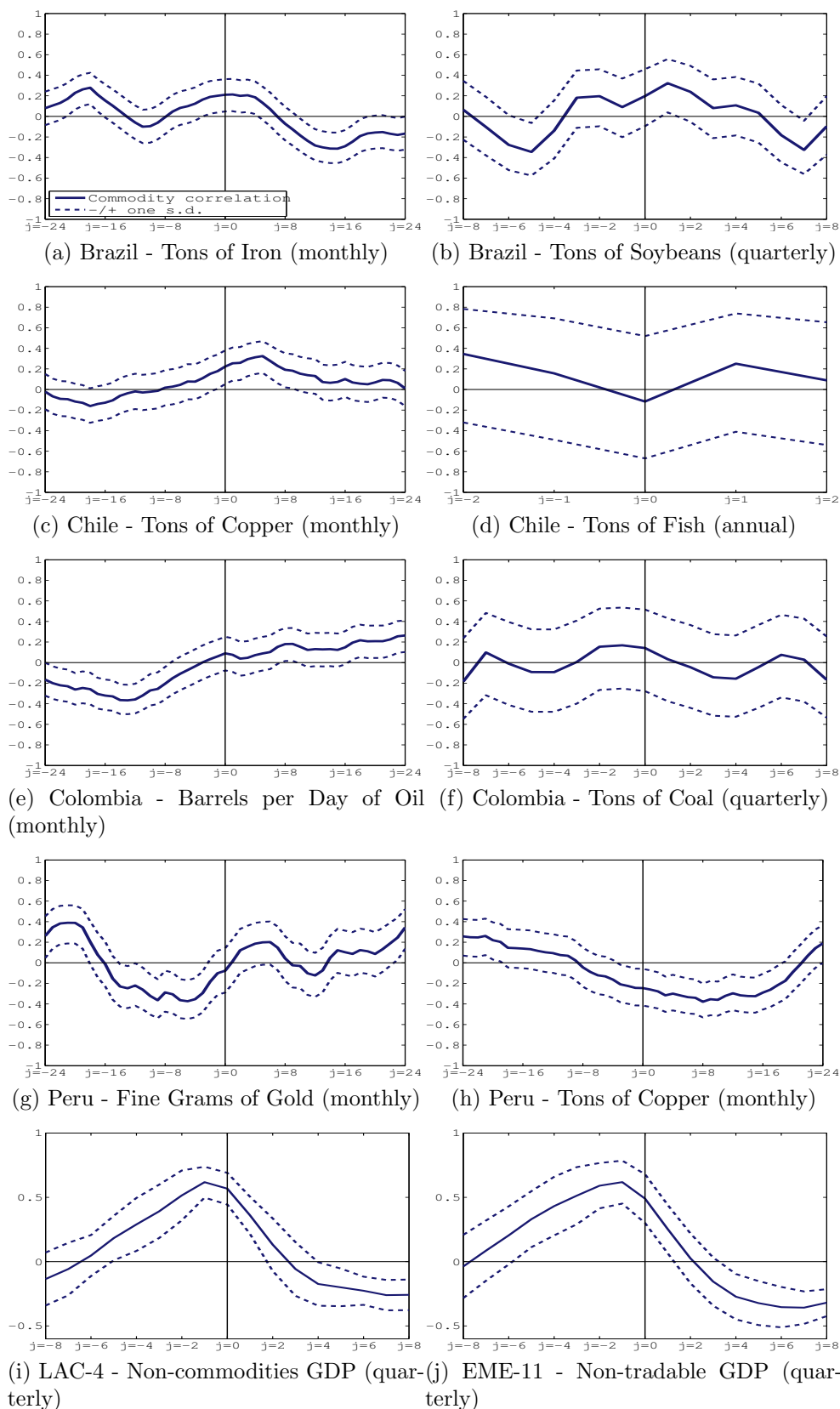
3636.7

(a) Baseline

(b) Non-Stationary Commodity Prices

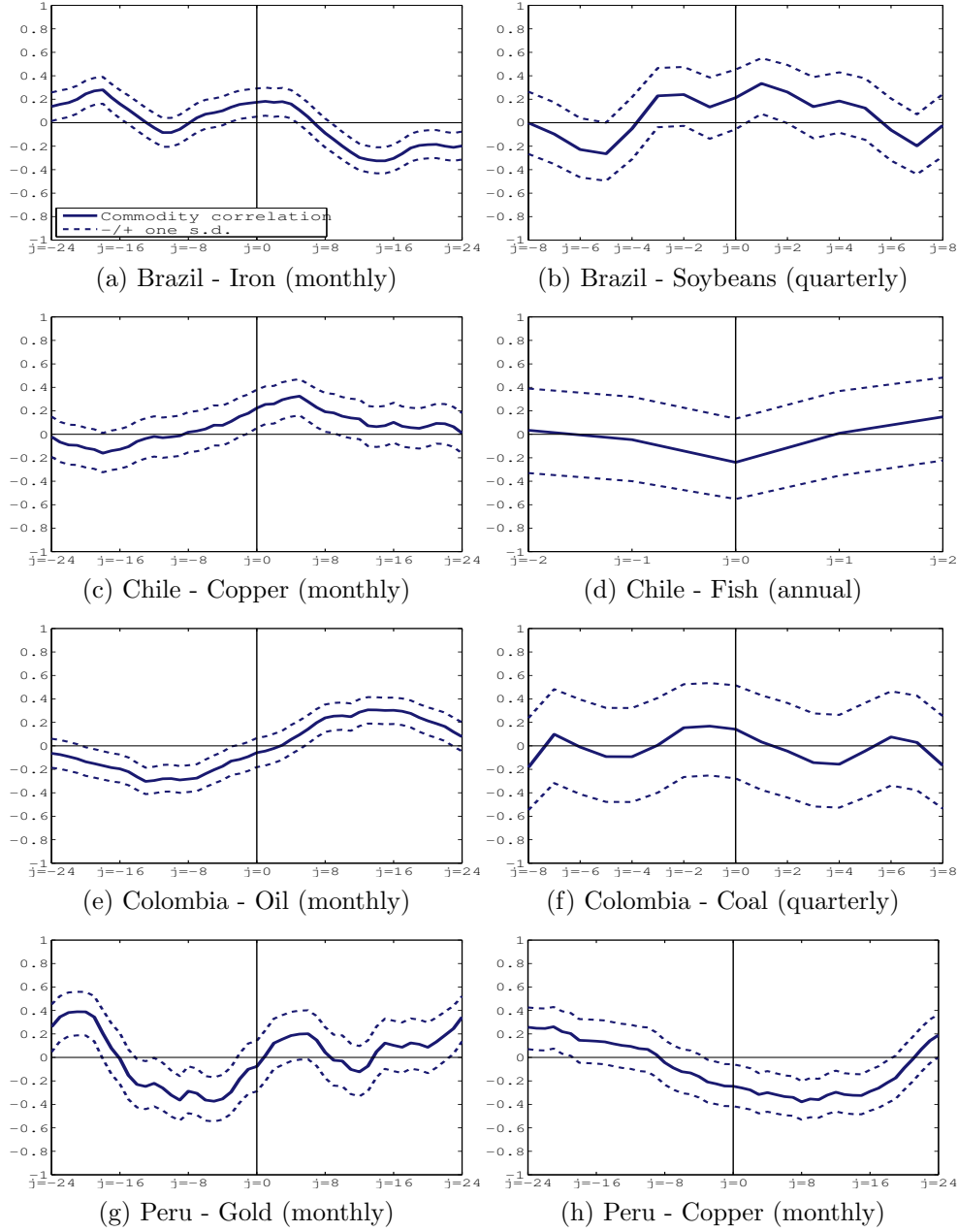
Note: The panels report the forecast error variance decomposition (FEVD) of real income calculated at the posterior mean for an alternative model assuming non-stationary commodity prices. Panel (a): Baseline; Panel (b): Non-Stationary Commodity Prices. Marginal Likelihood are computed with Geweke's modified harmonic mean.

Figure A14. Correlation of Commodity and Non-Commodity Production and International Commodity Prices



Note: Panels (a) to (h) report the serial correlations between the volumetric production of commodities in each of the two main commodities of LAC4 countries in t and the international market price of each commodity in $t+j$ with $j = -4, \dots, 4$. The correlations were computed with the highest frequency available which is reported in the subtitle. Panels (i) and (j) present averages of the correlations between the non-commodity GDP and non-tradable GDP in t , respectively, with country-specific commodity price indices across LAC4 and EME11 $t+j$ with $j = -4, \dots, 4$. Sources: National Statistical Agencies and author's calculations.

Figure A15. Correlation of Commodity and Non-Commodity Production and International Commodity Prices (whole sample)



Note: All available sample when computing the correlation coefficients. Panels (a) to (h) report the serial correlations between the volumetric production of commodities in each of the two main commodities of LAC4 countries in t and the international market price of each commodity in $t + j$ with $j = -4, \dots, 4$. The correlations were computed using the highest frequency available; see subtitles for details. Sources: National Statistical Agencies and author's calculations.

Table A30. Granger Causality Tests			
Country	Commodity	Price to Quantity	Quantity to Price
Brazil	Iron (monthly)	0.196	0.501
	Soybean (quarterly)	0.725	0.709
Chile	Copper (monthly)	0.512	0.681
	Fish (annual)	0.271	0.895
Colombia	Oil (monthly)	0.399	0.706
	Coal (quarterly)	0.192	0.769
Peru	Copper (monthly)	0.792	0.683
	Gold (monthly)	0.058	0.062

Note: This table reports the p-value of the Granger causality test after estimating a two-variable VAR for every commodity and running a Granger test to determine whether the commodity price Granger causes the commodity production and conversely. The lag length was selected using the Schwarz criterion with maximum lag ten, four and two for monthly, quarterly and annual frequency, respectively.